

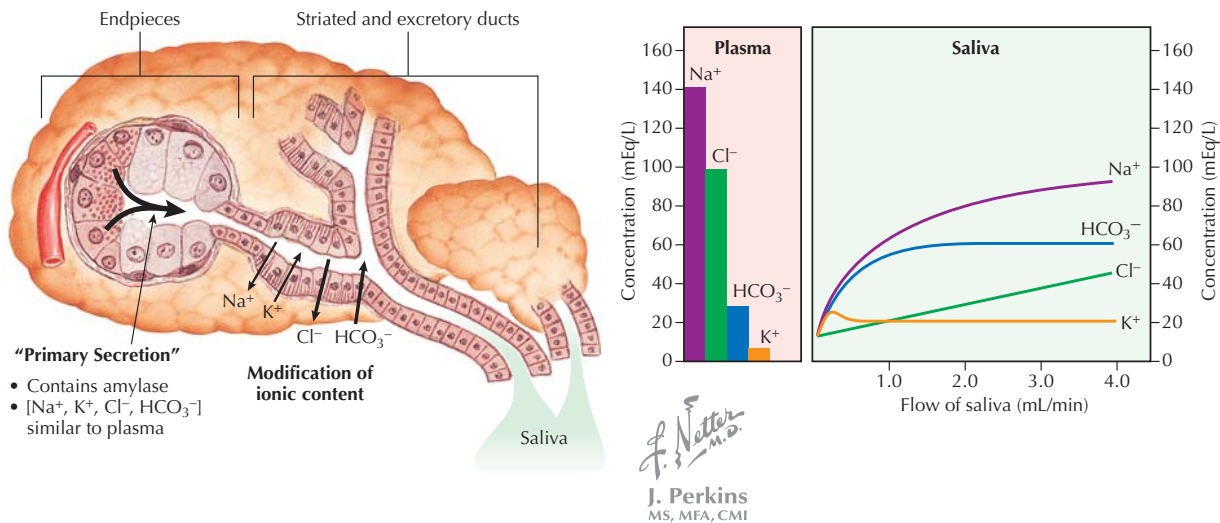


Figure 23.3 Salivary Acinar Cells During salivary secretion, blood flow to the acini is increased by parasympathetic stimulation, and ultrafiltrate from plasma (mostly serous fluid) enters the acini. Filtrate from the cells enters the lumen of the acinar cells, mixing with secreted mucus and α -amylase, creating the primary secretion. This secretion is modified as it passes through the ducts into the mouth. Lingual lipase (secreted from the Von Ebner's glands of the tongue) is added to the saliva in the mouth. The graph illustrates the effect of increasing salivary secretion on composition of saliva. The changes are due to the reduced modification of saliva through the ducts as flow increases.

of gastric juices with food produces **chyme**. The gastric pit is made of a variety of cells (Fig. 23.4):

- Superficial epithelial cells or surface mucous cells are formed from undifferentiated cells in the pit, then migrate to the surface. They secrete mucus and bicarbonate (HCO_3^-), which protect the gastric epithelium from the corrosive acid in the lumen, and are replaced every few days.
- Mucous neck cells protect the gastric mucosa by secreting mucus.
- Stem/regenerative cells differentiate to replace other cells in the pit.
- Parietal cells secrete HCl and intrinsic factor.
- Chief cells secrete pepsinogens and gastric lipase.
- Endocrine cells secrete somatostatin, and mast cells secrete histamine.

Because of the harsh physical environment, cells lining the stomach (and intestines) are continually replaced. This requires substantial energy expenditure, as well as a constant supply of proteins and lipids to make new cells.

Secretions from the Pit Cells

The gastric secretions primarily serve to digest the food; however, they also contribute to the regulation of acid secretion, motility, and protecting Vitamin B12. The gastric secretions consist of:

- HCl: Digests food, kills ingested bacteria, converts inactive pepsinogens to pepsins.

- Intrinsic factor: Essential gastric secretion required for B12 absorption in the terminal ileum.
- Pepsinogens: The inactive form of pepsins, which are proteases that are activated in the acid environment. Thus, protein digestion begins in the stomach.
- Gastrin: A gut hormone secreted from G-cells located in the antrum of the stomach and the duodenum. Gastrin stimulates HCl secretion and gastric motility (mixing), as well as lower GI motility (mass movements).
- Lipase (gastric lipase): Continues process of lipid digestion.
- Mucus: Thick mucus is secreted with HCO_3^- . The HCO_3^- remains trapped in the mucus layer at the epithelial cell surface, effectively buffering the cells from the acid environment in the lumen (Fig. 23.5).
- Other factors: Somatostatin (SS) inhibits HCl secretion; histamine stimulates HCl secretion.

In general, factors that stimulate gastric activity (e.g., vagus, gastrin) elicit secretion from chief, parietal, and mucus cells.

Production and Regulation of Gastric HCl

Acid secretion is important for the indiscriminate digestion of ingested matter. However, because of its corrosive nature, there are multiple factors that regulate its secretion from **parietal cells**. One factor that keeps HCl secretion under control is parietal cell morphology. Stimulated parietal cells are well organized, with tubulovesicles fusing with the plasma membrane to form intricate canaliculi that increase surface area for secretion of H^+ . In contrast, unstimulated

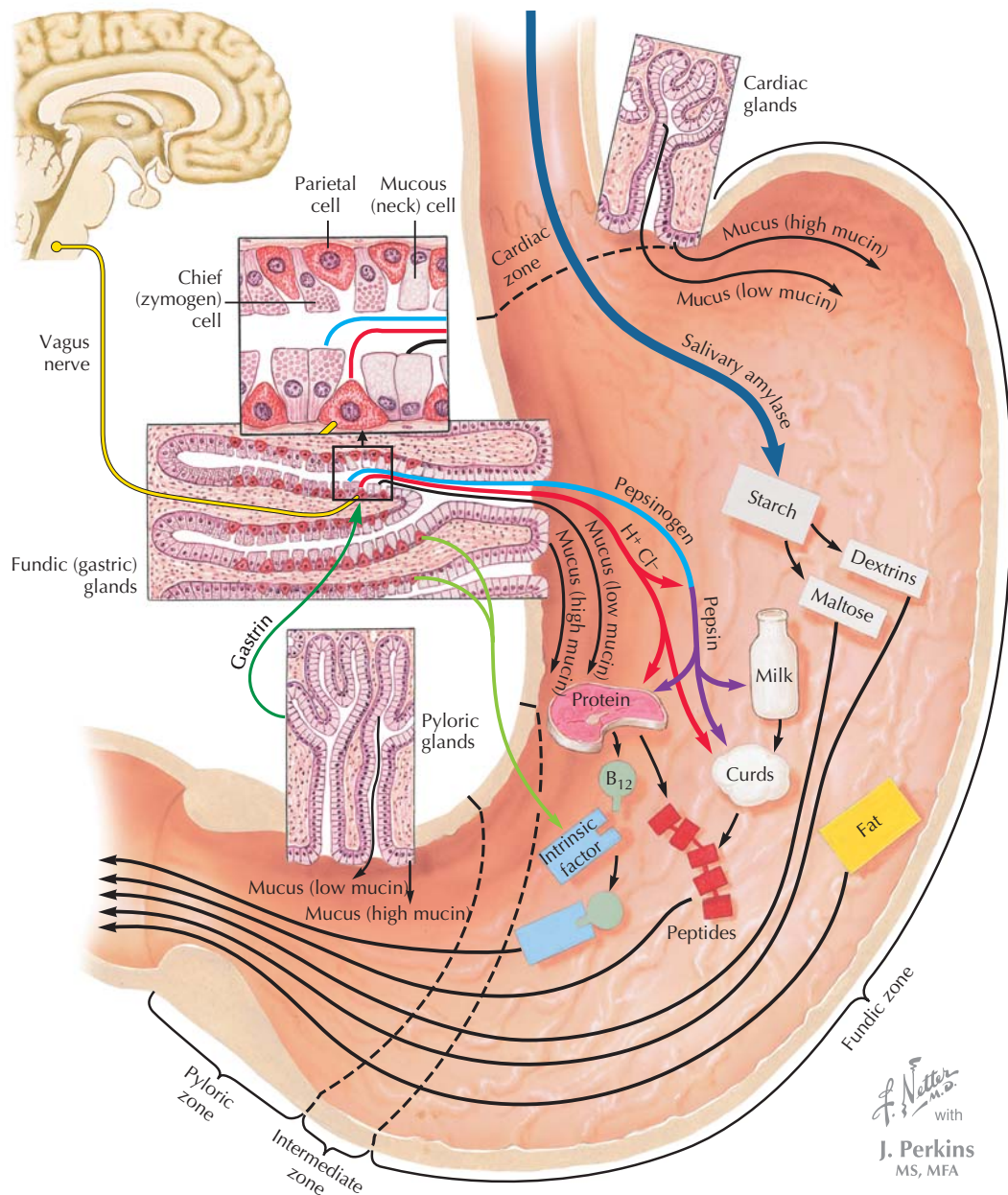


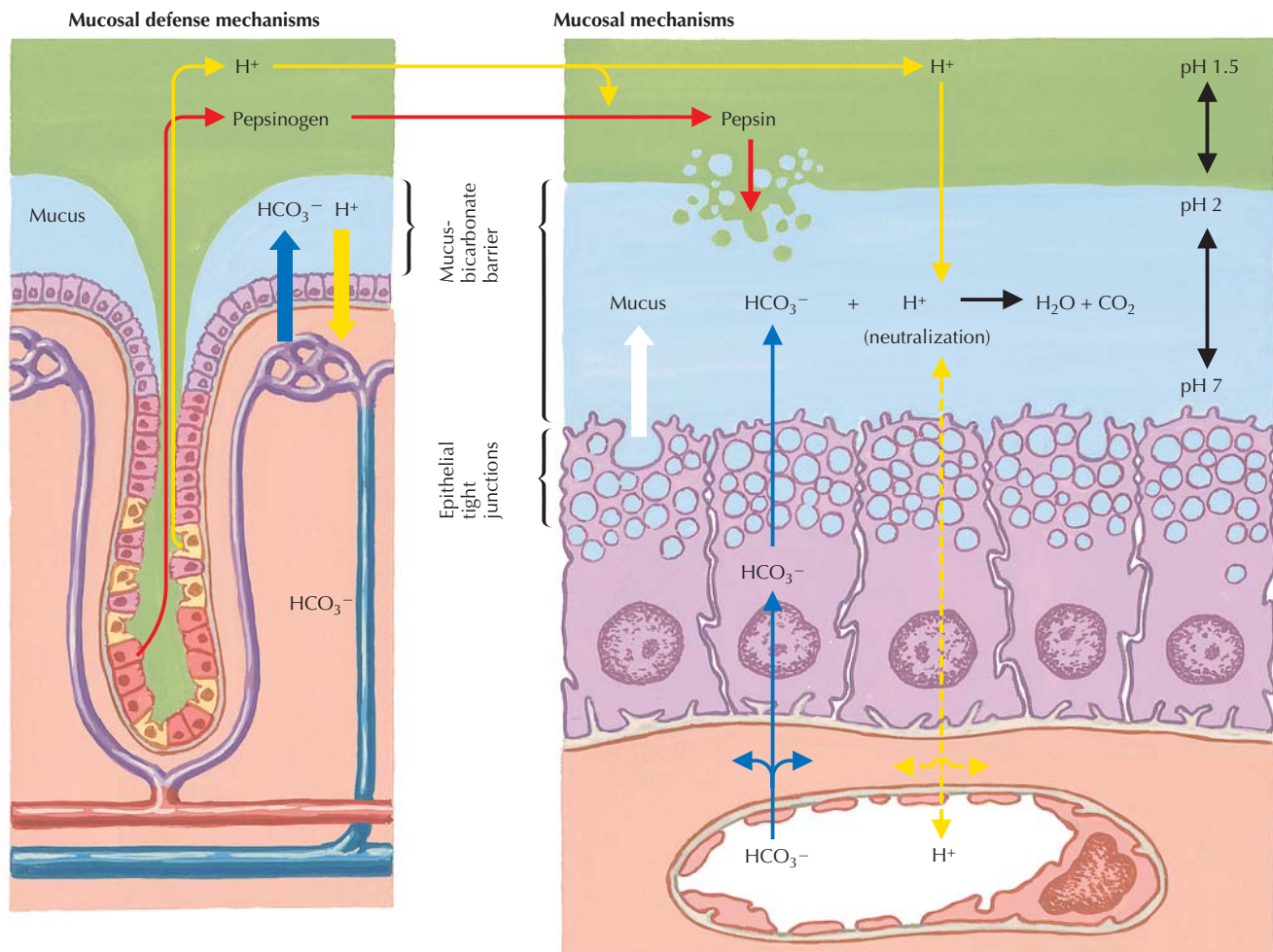
Figure 23.4 Digestion in the Stomach The stomach secretes concentrated HCl, which breaks ingested food into smaller fragment and activates the enzymes pepsin and lipase, to digest proteins and fats, respectively. The parietal cells, which secrete HCl, also secrete the protein intrinsic factor, which binds to vitamin B12 to prevent the vitamin's digestion by pancreatic proteases in the small intestine. In general, about 25% of nutrient digestion occurs before the chyme reaches the duodenum.

cells have less surface area exposed to the luminal surface of the gastric pits, keeping potential H^+ secretion minimal (Fig. 23.6).

Parietal cells secrete H^+ against a million-fold concentration gradient in the lumen. The gradient is maintained by **tight junctions**, which prevent H^+ leak back into the blood. The transport of H^+ against the gradient is accomplished by the

apical H^+/K^+ ATPase, or **proton pump**. This active antiport pump is the rate-limiting step in H^+ transport and is the site of regulation of acid secretion. Key aspects of acid production and secretion include the following:

- The production of H^+ from intracellular conversion of CO_2 to carbonic acid in the reaction ($CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3^-$) (carbonic anhydrase is required



Gastric mucosa and submucosa protected from chemical injury by mucus-bicarbonate surface barrier that neutralizes gastric H^+ and by epithelial "tight junctions" that prevent H^+ access to subepithelial tissue

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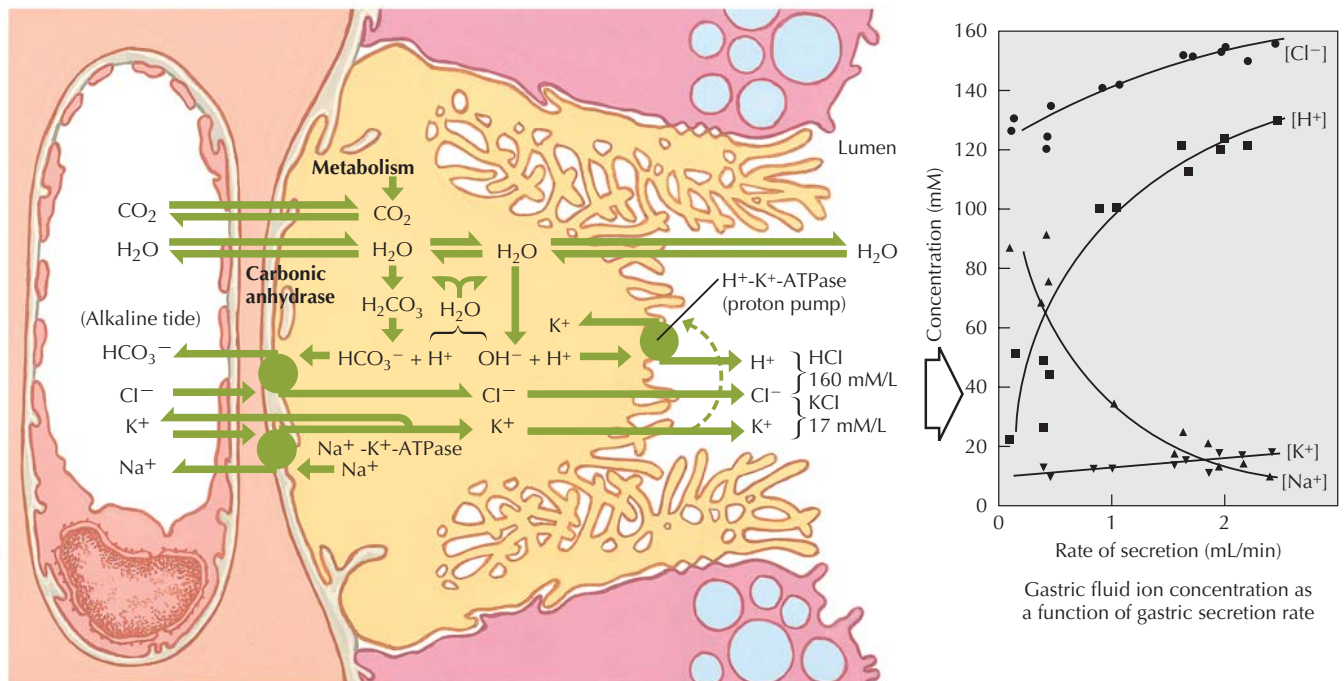
Figure 23.5 Protection of Gastric Mucosa Thick mucus from the surface epithelial cells traps the secreted bicarbonate on the surface of the gastric mucosa. The bicarbonate-containing mucus layer protects the cells from the caustic acid (pH ~1.5), keeping the pH at the cell surface ~7.0.

for this conversion). H^+ is then available for secretion by the proton pump.

- The entry of chloride into the cells via the $\text{HCO}_3^-/\text{Cl}^-$ exchanger at the basolateral membrane, and diffusion of chloride into the lumen of the pit through the apical membrane (down its concentration gradient).
- The exit of HCO_3^- from the cell into the blood via basolateral $\text{HCO}_3^-/\text{Cl}^-$ exchangers. In the actively secreting parietal cell, the large amount of HCO_3^- entering the blood is described as an **alkaline tide**.
- The movement of potassium back into the lumen with Cl^- .

This process primarily produces HCl in the lumen but also produces some KCl . Regulation of acid production takes place through stimulation and inhibition of the **proton pump**, which is the target for most therapeutic interventions to reduce acid secretion (for gastroesophageal reflux disease [GERD], ulcers, etc). Proton pump inhibitors (PPI) such as omeprazole and histamine H_2 blockers (ranitidine, cimetidine) are used for this purpose.

Constituents of the gastric juices vary with the rate of secretion (see Fig. 23.6). Quiescent, nonstimulated cells produce fluid at a low rate; the fluid contains higher levels of Na^+ and Cl^- (more bufferlike) and low concentrations of H^+ .



Parietal cell mechanisms of acid (H⁺) secretion involve series of chemical exchanges across basal membrane, with final active exchange of H⁺ for K⁺ mediated across apical (secretory) membrane by H⁺-K⁺ ATPase (proton pump).

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Figure 23.6 Parietal Cell HCl Production The apical H⁺/K⁺ ATPase (proton pump) causes the active secretion of H⁺ into the lumen of the gastric pits. This combines with secreted Cl⁻ to form concentrated HCl. HCl production is stimulated by vagal efferent fibers, gastrin, and histamine, which are active when there is food in the stomach. The H⁺/K⁺ ATPase is the target for proton pump inhibitors (PPI) to decrease gastric acid release in patients with GERD or ulcers.

However, when cells are stimulated, Na⁺ concentrations drop and H⁺ concentrations increase in the juice, producing more HCl.

Regulation of Gastric Acid Secretion

There are three phases of gastric acid secretion:

- **Cephalic phase:** Food in the mouth initiates vagal release of acetylcholine (ACh) at the parietal cell, stimulating acid secretion. Vagal fibers also terminate on enterochromaffin-like (ECL) cells and G-cells, stimulating histamine and gastrin secretion, respectively; histamine and gastrin promote acid secretion.
- **Gastric phase:** Stretch, initiated by food in the stomach, releases gastrin from antral G-cells. The gastrin in the bloodstream circulates back to the stomach, stimulating the proton pump of the parietal cells. At the same time, the vagus also stimulates the proton pump directly, as well as by increasing gastrin-releasing peptide (GRP) and histamine, which also act on the pump.
- **Intestinal phase:** When chyme enters the duodenum and jejunum, hormones are released that feed back to regulate acid secretion (see following).

Several factors **stimulate** acid secretion:

- The *vagus* acts directly at parietal cells and indirectly through stimulation of gastrin and histamine release and inhibition of local somatostatin release (Fig. 23.7).
- *Histamine*, from ECL cells, diffuses through the mucosa to act on adjacent parietal cells.
- *Gastrin*, carried through the blood, acts directly on parietal cells.
- *Insulin*, carried through the blood, acts directly on parietal cells and promotes HCl secretion.
- *Caffeine* (a phosphodiesterase inhibitor) increases cAMP in the parietal cells, increasing proton pump activity.
- *Stress*, although not well understood, appears to increase acid secretion in certain people and may be a cofactor in ulcer formation (with *Helicobacter pylori*).

Several factors **inhibit** acid secretion:

- *Somatostatin* (SS), released from endocrine cells in the gastric pit, acts in a paracrine manner on the parietal cells, as well as on G-cells to inhibit gastrin.

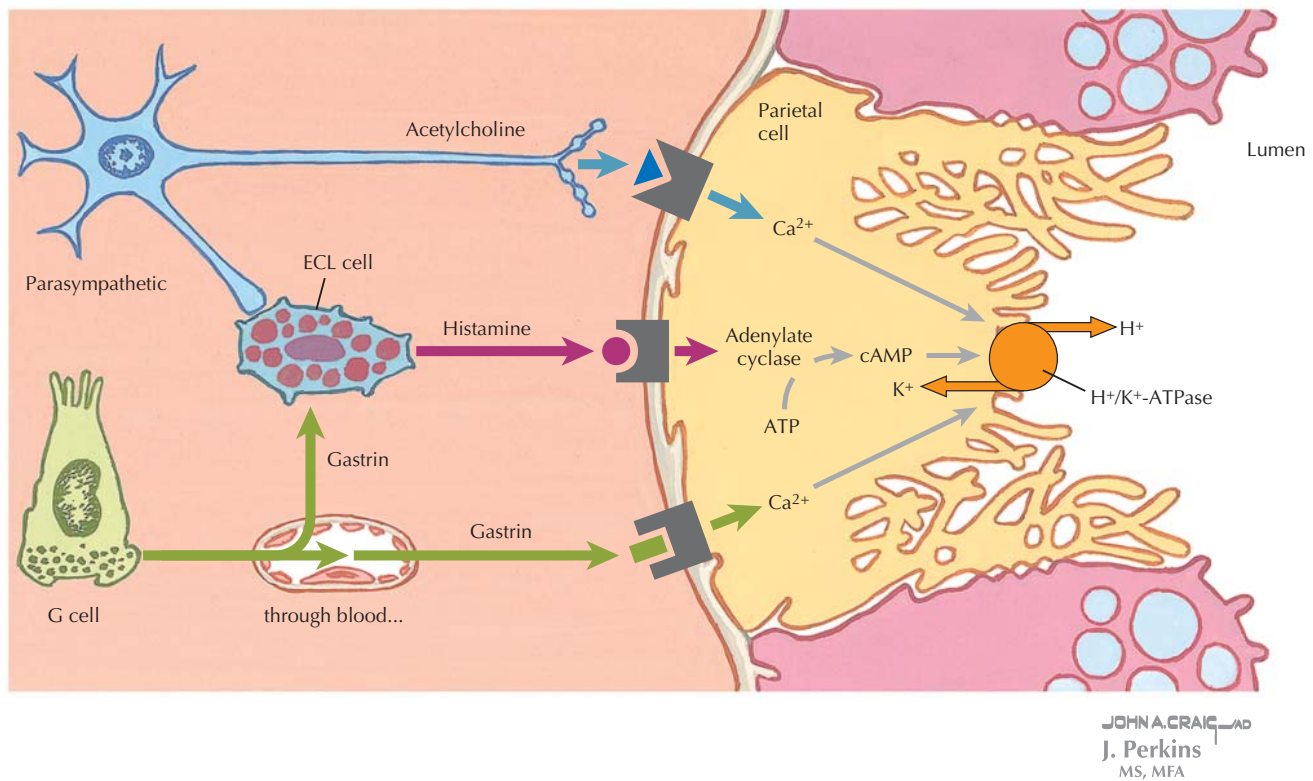


Figure 23.7 Signal Transduction Mechanisms Regulating HCl Secretion Stimulation of the proton pump can be achieved by both calcium and cAMP-mediated messenger systems.

- *Glucose insulinotropic peptide* or *gastric inhibitory peptide* (GIP), released from the duodenum and jejunum, acts directly on the parietal cells.
- *Secretin*, released from the duodenum and jejunum, acts at the G-cells to suppress gastrin.
- *Peptide YY*, released from various areas of the GI tract in response to fats, helps shut off acid and pancreatic secretions when chyme is leaving the upper GI tract. The effects on gastric acid may be through suppression of ACh release from cholinergic fibers, and/or stimulation of paracrine somatostatin.

While these factors regulate acid secretion, it is important to understand that they are not all-or-nothing events—for the most part they work in concert to modulate the amount of acid being secreted during the digestive period. Thus, for example, while the vagus, gastrin, and histamine are stimulating acid secretion, secretin works to moderate gastrin release, and somatostatin and GIP act on the parietal cells, keeping acid secretion at a reasonable rate. The dynamic nature of the system is critical to proper function. One reason that potentiation of acid secretion can occur is that various factors use different second messenger systems (see Fig. 23.7).

SECRETIONS FROM THE SMALL INTESTINE: BUFFERS, ENZYMES, AND HORMONES

The crypts of Lieberkühn, located at the bottom of the villi in the small intestine, have a variety of cell types that produce and secrete buffers, enzymes, and hormones that facilitate digestion and absorption of the nutrients in the chyme (Fig. 23.8). **Brunner's glands** are located in the duodenum and secrete thick mucus, which helps to protect the early part of the small intestine from the acidic chyme leaving the stomach. These glands are stimulated by secretin and activity of the vagus. The **Paneth cells** are deep in the crypts and are stimulated by secretin to secrete ions and water, which buffer the chyme. They also secrete **lysozyme**, which has antimicrobial actions. **Goblet cells** are located throughout the small intestine and secrete mucus. Cells also release the enzyme **enterokinase**, which cleaves the inactive zymogen trypsinogen, forming the active enzyme trypsin. This occurs in the duodenum, activating the powerful protease in the presence of chyme.

The small intestine produces and secretes a variety of endocrine hormones that regulate digestion and motility (Table 23.3; the major actions of the hormones are in bold). The effects on digestion are detailed in Chapter 26.

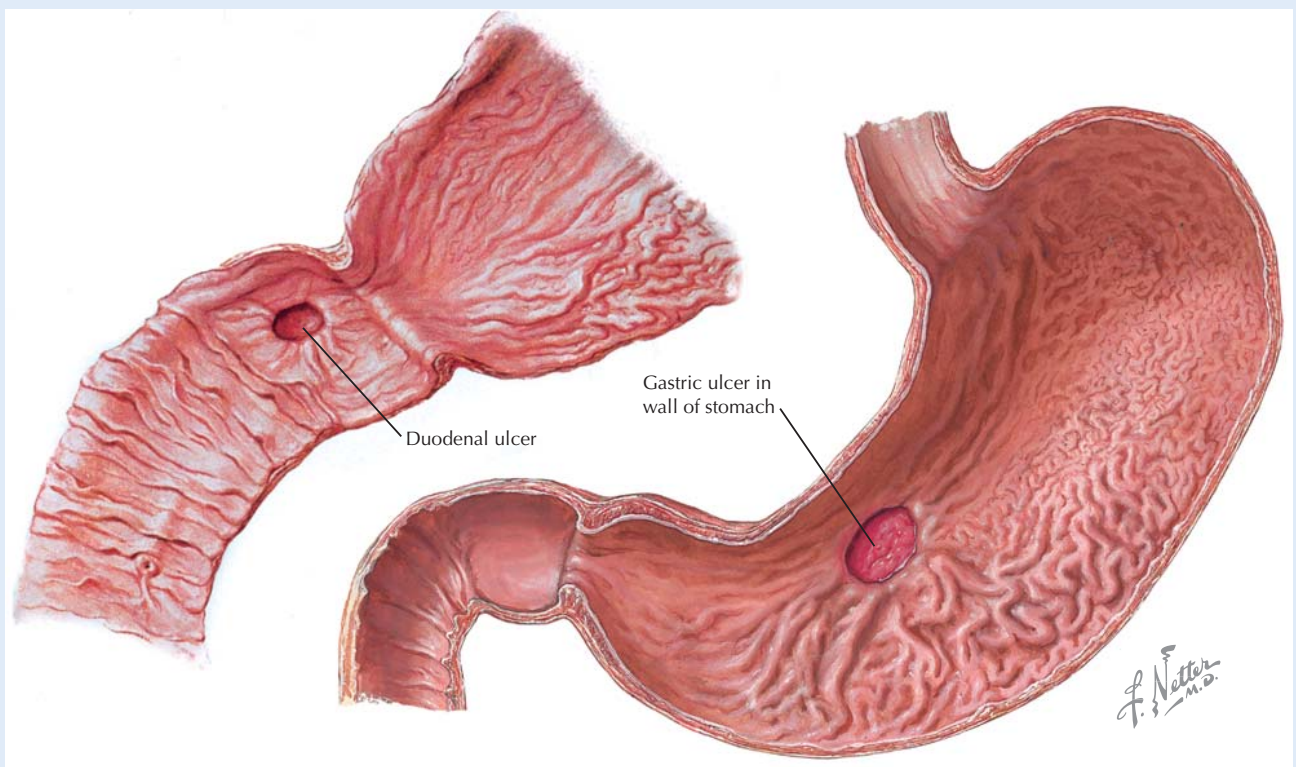
CLINICAL CORRELATE

Ulcers

Ulcerations were thought to result mainly from ingestion of NSAIDs (nonsteroidal anti-inflammatory drugs such as aspirin) and stress until about 25 years ago, when Australian investigators discovered *Helicobacter pylori* residing in the gastric and duodenal mucosa of patients with gastric and duodenal ulcers. It is now known that *H. pylori* accounts for almost all non-NSAID-related gastric ulcers and nearly all duodenal ulcers. How can the bacteria live in the acidic (pH 2) environment of the stomach? *H. pylori* is resistant to HCl and can burrow into the mucosal layer because it secretes **urease**, which breaks urea into CO_2 + ammonia (NH_3). This disrupts the bicarbonate–mucus layer protecting the gastric mucosa and allows acid and pepsins to infiltrate the mucosa, initiating ulcerations. Interestingly, while most people

have the bacteria (fecal-oral contamination from food or water), a critical number of bacteria must be present to initiate enough **urease** production to disrupt the mucosa and expose the epithelium to ulceration. Treatment is with antibiotics (long term) and proton pump inhibitors (PPIs).

NSAIDs, including aspirin, ibuprofen, and similar drugs, can also cause gastric ulcers. Because prostaglandins are cytoprotective (by decreasing parietal cell acid production and increasing mucus production), the suppression of prostaglandins (especially PGE_2) by NSAIDs can lead to ulcerations. The ulcerative effect is enhanced when the drugs are taken on an empty stomach or in the presence of caffeine, which stimulates acid secretion. Stress, coffee, and NSAIDs (particularly when taken on an empty stomach) may have additive effects in aggravating existing ulcers. Treatment is with PPIs (and behavior modification).



Ulcerations of Stomach and Duodenum

PANCREATIC BUFFER AND ENZYME SECRETIONS

The pancreas synthesizes and releases both endocrine secretions (into blood) and exocrine secretions (through ducts). The exocrine secretions serve to help buffer the acidic chyme and are crucial for the proper digestion of chyme, because

approximately 70% to 75% of digestion occurs in the proximal small intestine by the pancreatic enzymes. The exocrine secretions exit the pancreatic ducts into the common bile duct, mixing with liver and gallbladder secretions. The bile duct enters the duodenum through the **sphincter of Oddi**, at the **papilla of Vater** (Fig. 23.9A).

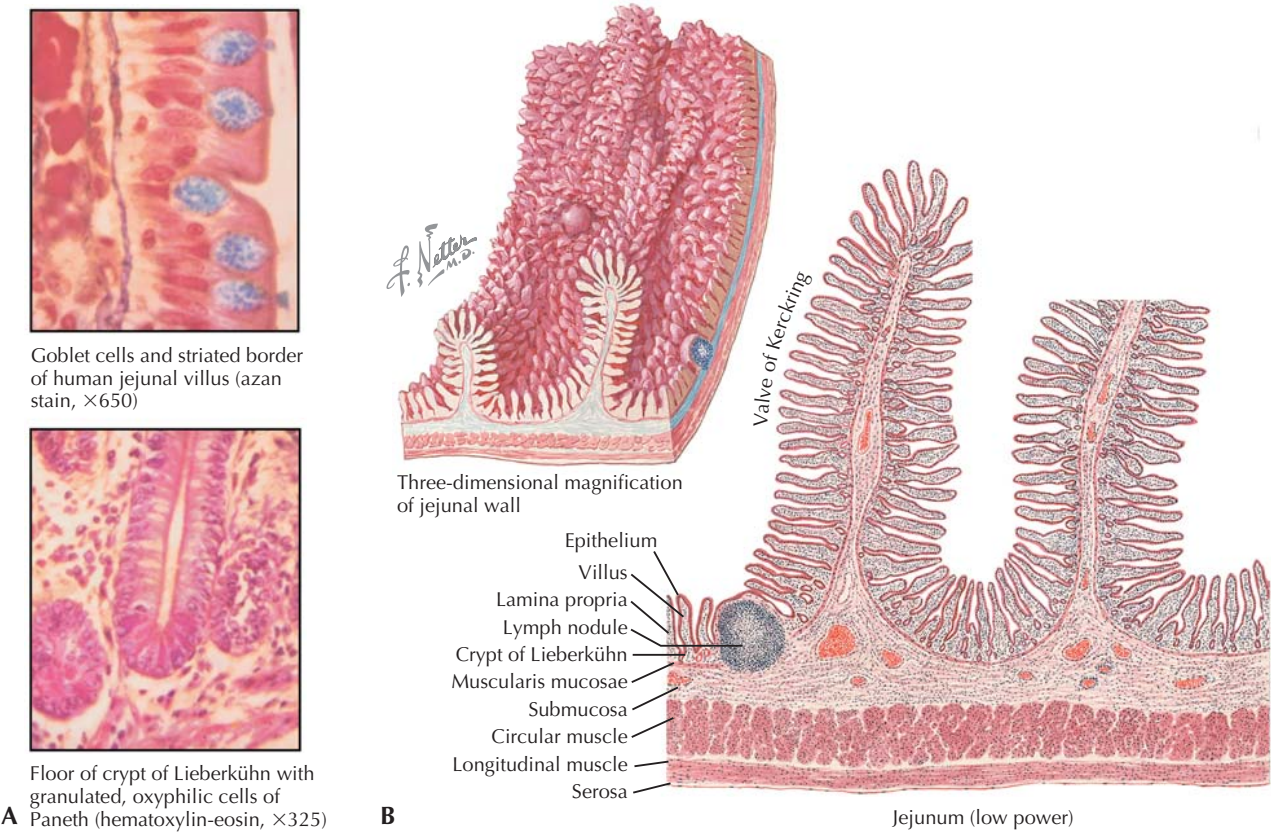


Figure 23.8 Crypts of Lieberkühn and Cell Types A, Micrographs of goblet and Paneth cells in small intestinal mucosa. B, Brush border lining of the small intestine, illustrating crypts of Lieberkühn within villus projections.

Table 23.3 Major GI Hormones

GI Hormones	Site of Secretion	Primary Stimuli	General Actions
Gastrin	G-cells in antrum of stomach and duodenum	Stretch, peptides and amino acids, vagus (through GRP)	↑gastric H ⁺ ↑gastric mixing ↑lower GI tract motility
Secretin	S-cells of the duodenum	Acidic chyme	↑pancreatic buffer (HCO ₃ ⁻) secretion ↑biliary and small intestine buffer secretion ↓gastric H ⁺ (by ↓gastrin) ↓gastric emptying
Cholecystokinin (CCK)	I-cells of the duodenum and jejunum	Small peptides and amino acids, fats	↑pancreatic enzyme secretion contracts gallbladder and relaxes sphincter of Oddi ↑pancreatic and biliary buffer secretion ↓gastric emptying ↑lower GI tract motility
Gastric inhibitory peptide, aka glucose insulinotropic peptide (GIP)	Duodenum and jejunum	Fatty acids, glucose, amino acids	↓gastric H ⁺ secretion ↑pancreatic insulin secretion ↓gastric emptying
Motilin	Mo cells of the duodenum	Fasting	↑phase III contractions of the MMC

GRP, gastrin-releasing peptide; MMC, migrating myoelectric complex.
(Reprinted with permission from Hansen J: Netter's Atlas of Human Physiology, Philadelphia, Elsevier, 2002.)

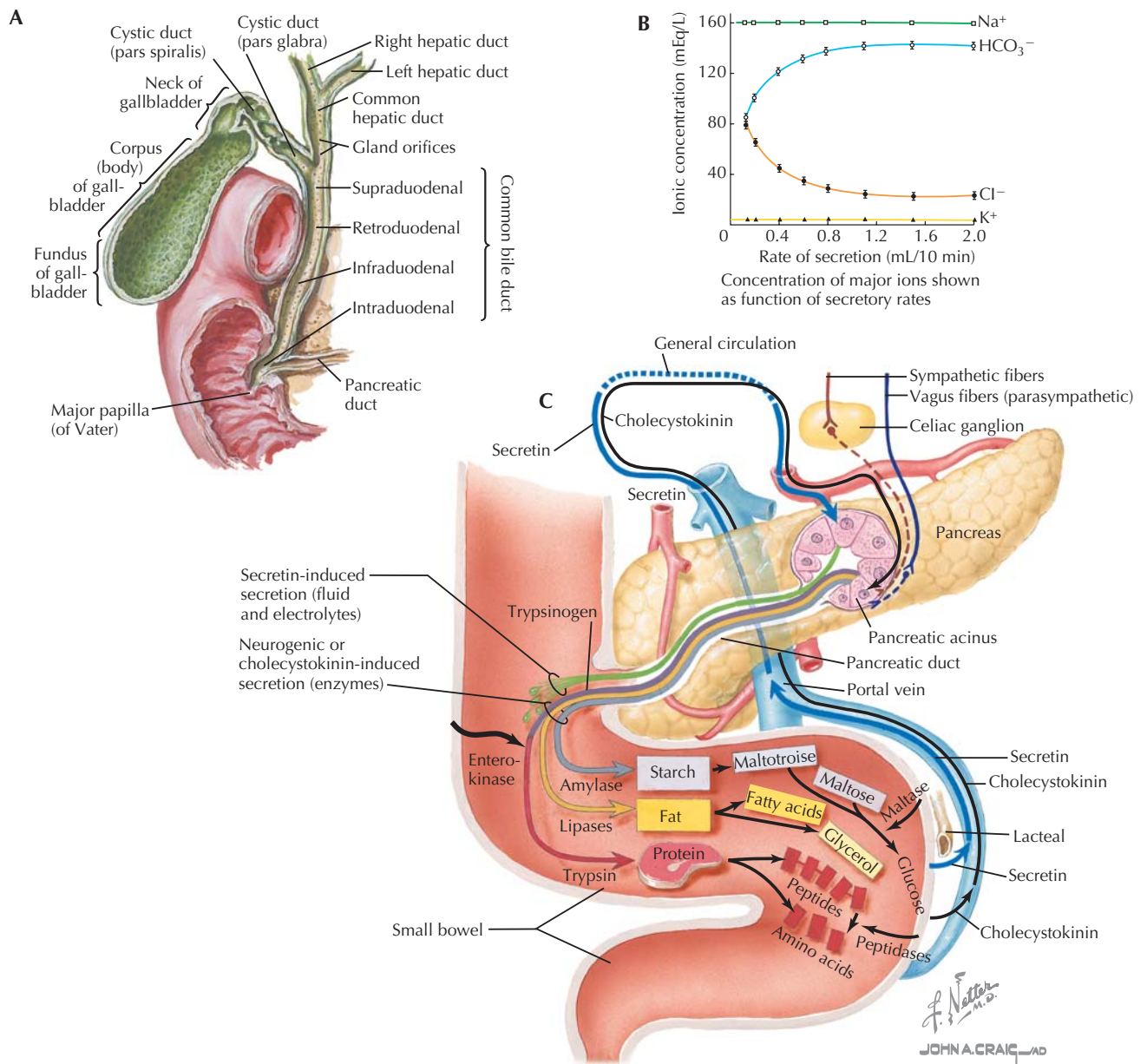


Figure 23.9 Structure and Secretion of the Pancreas A, Pancreatic, liver, and gallbladder secretions flow through the common bile duct and into the duodenum through the sphincter of Oddi at the papilla of Vater. B, As the rate of pancreatic secretions increases, the concentration of electrolytes changes, producing high sodium bicarbonate buffer. C, In response to food/chyme in the stomach and duodenum, vagal efferents and hormones stimulate pancreatic buffer and enzyme secretions.

Secretin, released in to the blood from duodenal S-cells in response to acidic chyme, binds to its receptors on the pancreatic centroacinar cells and stimulates **electrolyte and fluid secretion** (for buffering) (see Fig. 23.9C). As with the salivary glands, as the secretion goes through the ducts, there is modification of electrolytes. As the rate of secretion increases, the composition of the fluid changes, becoming rich in HCO_3^- (see Fig. 23.9B).

Cholecystikinin (CCK) is released into the blood from the duodenal I-cells and acts on the pancreatic acinar cells. Along with the vagus, CCK stimulates **pancreatic enzyme secretions**, which include:

- **Pancreatic proteases:** Trypsin, chymotrypsin, and elastase, the major proteases, are stored and released as inactive zymogens (trypsinogen, chymotrypsinogen,

and proelastase). This protects the pancreas and ducts from digestion. When the zymogens enter the duodenum, **enterokinase** (in the duodenal brush border) cleaves trypsinogen to trypsin, and then trypsin can activate additional trypsinogen (autoactivation), as well as the other proteases.

- **Pancreatic α -amylase:** Starch digestion begins in the mouth with salivary α -amylase. Digestion continues in the small intestine by pancreatic α -amylase, which hydrolyzes the starch molecules to maltose, maltotriose, and isomaltose. The amylase is activated by Cl^- .
- **Pancreatic lipase and colipase:** Pancreatic lipase hydrolyzes triglycerides to monoglycerides and free fatty acids; colipase is a cofactor for this process. Other lipases convert cholesterol esters to cholesterol and fatty acid, and phospholipids to lysophospholipids and fatty acid. However, pancreatic lipase cannot readily access the lipids in the small intestine because of the presence of bile. Colipase is released as precolipase and is activated in the duodenum by trypsin. The colipase binds to bile and allows the pancreatic lipase to access the lipids (see Chapter 25).

GALLBLADDER FUNCTION

The function of the gallbladder is to store bile. The sphincter of Oddi relaxes early in feeding, via vagal stimulation, releasing the contents of the bile duct (bile, electrolytes) into the duodenum. When chyme enters the duodenum, CCK is released. CCK opens the sphincter and causes gallbladder contraction. The stored bile enters the duodenum and begins lipid emulsification (see Chapter 25).

COLONIC SECRETIONS

In the colon, there is secretion of mucus, HCO_3^- , H^+ , and K^+ . The mucus is required to lubricate chyme as it is dehydrated and formed into feces. With colonic reabsorption of sodium

(through Na^+/H^+ exchangers), H^+ is secreted into the lumen. Cl^- is also absorbed, in part through luminal $\text{Cl}^-/\text{HCO}_3^-$ exchangers. Thus, H^+ and HCO_3^- are secreted into the lumen of the colon. In addition, a concentration gradient develops for K^+ , which then diffuses into the lumen. Potassium secretion can also occur when aldosterone is elevated (during volume contraction), because aldosterone increases Na^+/K^+ antiporters on the luminal membrane, reclaiming additional sodium in exchange for K^+ . Because of the relatively high HCO_3^- content of feces, chronic diarrhea can cause metabolic acidosis (see Chapter 20).

Gasses are formed throughout the GI tract, diffusing in and out of cells, and for the most part we are unaware of the fluxes. CO_2 is constantly produced, but in addition, bacterial action can produce H_2 , N_2 , and methane, for example. Typically, a person produces and expels (as flatus) about 200 milliliters (mL) of colonic gas each day. This can change, with diets high in carbohydrates and meat protein increasing gas production.

CLINICAL CORRELATE Zollinger-Ellison Syndrome

Gastrinomas are tumors (usually pancreatic) that secrete gastrin, but do not have any of the normal feedback systems that control gastrin release. These tumors cause **Zollinger-Ellison syndrome**, in which the constant secretion of gastrin causes uncontrolled, high levels of gastric HCl release and ulcerations of the stomach mucosa. In addition, the high acid secretion acidifies the duodenum and early jejunum, impeding the actions of digestive enzymes and precipitating bile. This especially affects lipid digestion and absorption, because the low duodenal pH denatures pancreatic lipase, significantly reducing lipid digestion. This results in steatorrhea and excess bile salts in the feces. Treatment is by removal of the tumor and the use of PPIs to further reduce acid secretion, allowing ulcerations to heal.

CHAPTER 24

Hepatobiliary Function

OVERVIEW OF LIVER FUNCTIONS

The liver has a myriad of functions that revolve around vascular, metabolic, and secretory aspects. The **hepatocytes** are the cells making up the parenchyma of the liver. The basic functions of the liver include:

- Regulation of carbohydrate, lipid, and protein metabolism: The liver receives newly absorbed nutrients through the portal vein, as well as contents of systemic blood, and processes them according to need. It produces albumin, fibrinogen, immunoglobulins, binding proteins, cholesterol, lipoproteins, and other important molecules.
- Regulation of cholesterol production and excretion: The body requires cholesterol, which the liver is able to synthesize.
- β -Oxidation of fatty acids: While many tissues use β -oxidation as an alternate energy source when glucose is not present, the liver has a high rate of β -oxidation during the interdigestive period.
- Bile acid production and secretion: Bile is necessary for efficient lipid absorption. Bile is polar, allowing it to incorporate lipids into a **micelle**, and then taxi the lipids through the unstirred water layer adjacent to the enterocytes (see Chapter 25). Without bile, the bulk of the hydrophobic lipids cannot get near the brush border.
- Endocrine functions: The hepatocytes produce and secrete hormones into blood, including insulin-like growth factor-1 (IGF-1), hepatocyte growth factor, angiotensinogen, and cytokines. It converts thyroxine (T₄) to active triiodothyronine (T₃) and participates in activation of vitamin D.
- Detoxification: The liver contains reticuloendothelial cells, the **Kupffer cells**, which are fixed macrophages in the endothelial lining of hepatic sinusoids. As blood passes through the liver, old and damaged erythrocytes undergo phagocytosis by the Kupffer cells. Hormones, drugs, and other chemicals are metabolized by the hepatocytes.
- Vitamin and iron storage: The liver stores several elements crucial to normal body functions (B12, folic acid, iron). Vitamins can be stored for weeks to months, and in the case of essential vitamins like B12 (cobalamin), this provides a source of the vitamin if the dietary supply is depleted. The liver is also the site of large iron stores,

bound to the protein **ferritin** (containing 25% of the body's iron). When needed, the iron (bound to transferrin) is released into the blood and enters the bone marrow for production of hemoglobin.

This great variety of functions makes the liver a crucial organ for maintaining immune cell function and proper blood glucose levels, excreting waste products, processing proteins, and even indirectly regulating blood pressure and fluid homeostasis (through synthesis of plasma proteins) (Fig. 24.1). When part of the liver is damaged, it has the ability to regenerate functional hepatocytes, and this compensatory action allows it to maintain adequate metabolic function. In addition, the ability of the liver to achieve a high level of metabolic activity is dependent on the blood flow.

LIVER STRUCTURE AND BLOOD FLOW

The liver is like a sponge. The hepatocytes are surrounded by sinusoidal capillaries and lymph vessels (Fig. 24.2). This allows for the free flow of blood to the cells, and the cellular products secreted into the perisinusoidal **space of Disse** can be released back into blood, or into the lymphatic vessels. **Kupffer cells** are affixed to the sinusoidal membranes and are present throughout the liver. These cells phagocytose bacteria and damaged red blood cells. The hepatocytes metabolize hormones and drugs and clear the waste, “detoxifying” the blood. The hepatocytes also produce bile and electrolyte solutions that drain into the **bile canaliculi**, which lead to the bile duct.

Under normal conditions, the liver contains ~400 milliliters (mL) of blood, which is ~8% of the total blood volume. This illustrates the “blood storage” aspect of the liver. Two key factors that make the system work properly are that hepatic vein pressure is very low, near 1 mm Hg, and that there is no obstruction of blood flow through the liver sinusoids. This allows the free flow of blood through and out of the liver into the vena cava. If there is obstruction of flow through the liver (cirrhosis, hepatitis) or increased hepatic vein pressure (congestive heart failure), blood will back up and pressure will rise, causing **portal hypertension**. The volume of blood associated with the liver can also increase to 1 liter (L), causing **hepatomegaly** (enlarged liver).

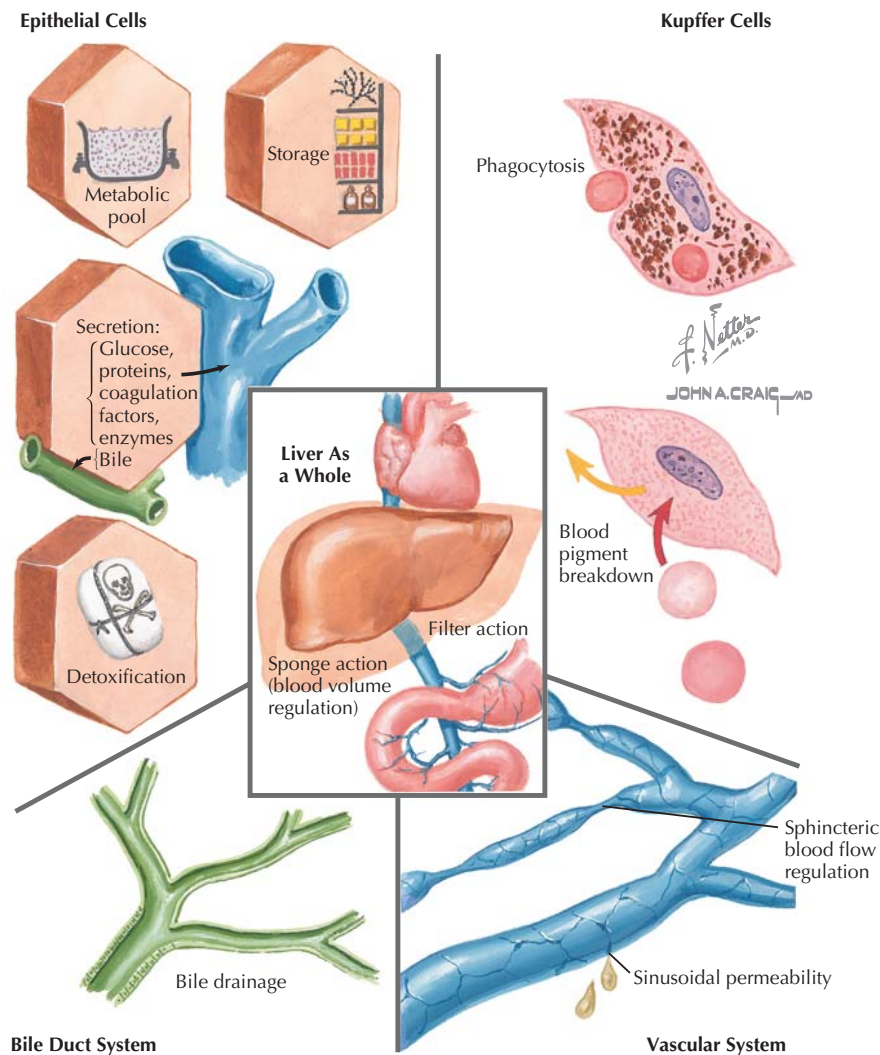


Figure 24.1 Overview of Liver Function The liver performs many functions, including metabolism of carbohydrates, proteins, and lipids; phagocytosis and removal of waste and bacteria; detoxification; synthesis of bile; blood volume control; and vitamin storage.

The blood supply to the liver comes from the hepatic artery and portal vein. The systemic blood from the hepatic artery enters the liver at a rate of ~450 mL per minute. The portal vein carries the blood from the intestines (~1 L per minute), and the arterial and venous blood intermingles in the sinusoidal capillaries. This amount represents about 30% of the cardiac output. As the blood from the portal vein enters the liver, nutrients, bacteria, and foreign bodies are processed—this is the “first pass” effect, which allows the absorbed materials to be “cleared” by the liver before the blood leaves through the hepatic vein into the systemic circulation. Although not all such substances will be cleared or metabolized in one pass, the bulk of the substances will be handled.



Ascites is fluid in the peritoneal cavity of the abdomen.

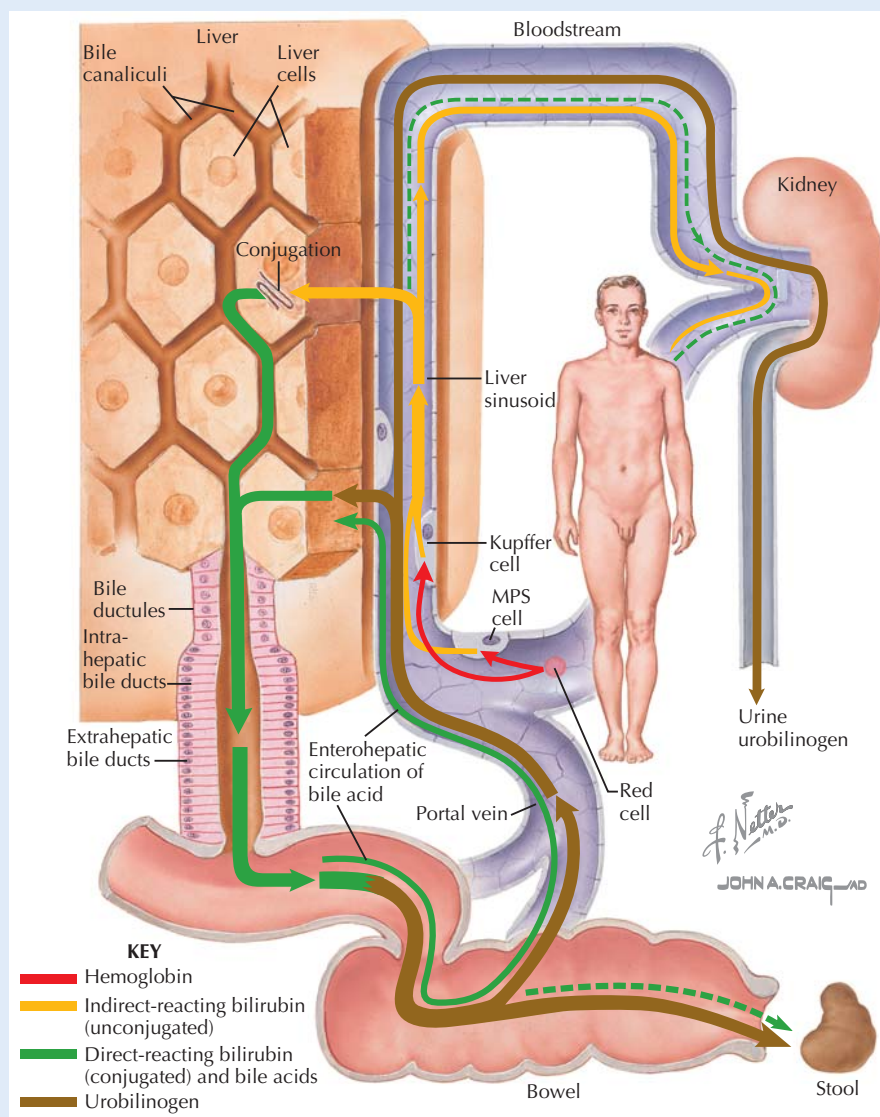
This can be caused by increased vascular pressure within the liver. Because the liver capillaries have a high permeability, when the pressure increases, fluid is forced out of the space of Disse and lymphatics, into the peritoneal space. Increased venous pressure, as in congestive heart failure, can cause ascites as well as peripheral **edema**. Edema results from elevated capillary hydrostatic pressure, increasing the Starling forces pushing fluid out of the capillaries into the interstitial space (in the lower extremities).

CLINICAL CORRELATE

Bilirubin and Jaundice

Red blood cells (RBCs) are viable for about 120 days, and new cells are continually being produced to replace damaged old cells. Phagocytic mononuclear cells throughout the body (especially Kupffer cells in the liver) remove and break down the old red blood cells, so that the iron can be recycled and waste eliminated. **Bilirubin** is a by-product of RBC degradation and is eliminated from the body by incorporation into bile (and subsequent loss in feces) and by urinary excretion as urobilinogen. The addition of bilirubin to bile, and its excretion in feces, contributes to the pigmentation of the feces. The bile pathway of excretion is important for ridding the body of excess bilirubin.

Jaundice occurs when plasma bilirubin is elevated. In patients it is observed as a yellowing tinge in the whites of the eyes and in skin tone and nail beds; stool is pale. **Primary jaundice** arises from hepatic dysfunction, as seen with obstructive liver disease (cirrhosis), blockage of bile ducts (by tumor or gallstones), or inflammation (hepatitis C). **Secondary jaundice** occurs from extrahepatic sources, such as abnormal lysis of RBCs (hemolytic disease). In most cases, when the primary cause of the jaundice is treated, the jaundice will abate as the excess bilirubin eventually clears (through bile and urinary excretion). An important distinction is that with secondary jaundice liver function is normal.



Bilirubin Production and Excretion

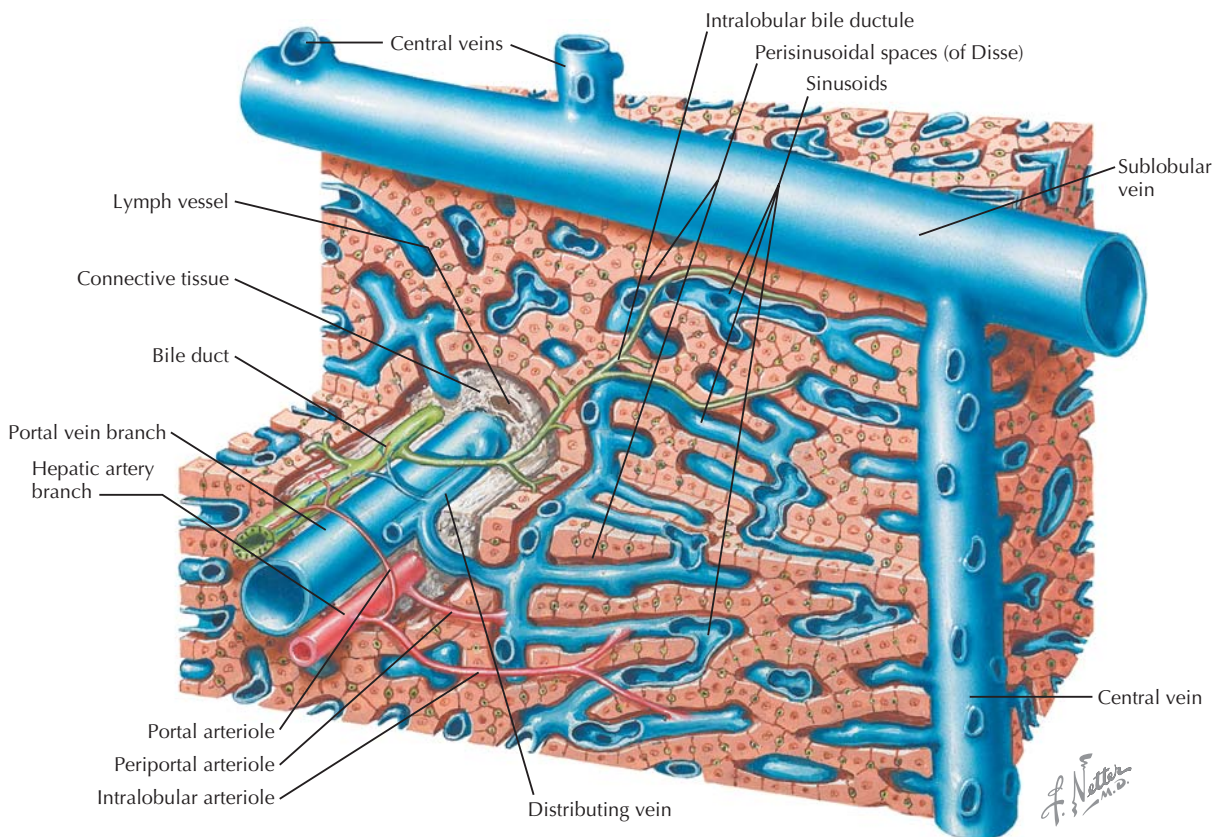


Figure 24.2 Structure of the Liver The liver is highly vascularized to accomplish its primary function of filtering portal and systemic blood. The capillary sinusoids surround the liver cells (hepatocytes), allowing efficient access of blood to the cells and transfer of products back into the blood. The sinusoids also contain the Kupffer cells, which are fixed macrophages that phagocytose damaged or aged red blood cells.

BASIC METABOLISM OF CARBOHYDRATES, LIPIDS, AND PROTEINS

Carbohydrates

The liver acts as a blood glucose monitor, storing glucose as glycogen and releasing it when blood levels are low. Carbohydrates are absorbed in the intestine as monosaccharides and are carried in the portal blood to the liver. Most of the glucose passes through the liver rapidly and is released into the systemic blood, where elevated insulin will facilitate its entry into tissues. In the liver, excess monosaccharides are handled by:

- **Conversion of other monosaccharides to glucose:** Fructose and galactose can be converted into glucose.
- **Glycogen synthesis and storage:** Excess glucose is polymerized and stored as glycogen. Stored glycogen can provide glucose for 17 to 24 hours during fasting. When blood glucose levels are low, glucagon (and other hyperglycemic hormones like epinephrine and growth hormone) stimulates glycolysis to break down glycogen



The ingestion of carbohydrates raises blood glucose levels; the **glycemic index** of a specific carbohydrate refers to the degree to which it raises blood glucose. Diets that produce high glycemic indices are believed to be associated with development of type II diabetes. High-fructose corn syrup is used in many foods, and because of its low glycemic index, has been recommended for people with diabetes. However, some studies suggest that elevated blood fructose levels may actually contribute to insulin resistance and hyperlipidemia, which, along with elevated blood pressure, are components of metabolic syndrome.

and release glucose into the blood. If the glycogen stores are not used, excess glucose (not released into the blood) will eventually be converted to triglycerides (TG) and transported to adipose tissue for storage (see “Lipids”).

- **Gluconeogenesis:** The liver (and to a lesser extent, the kidney) has the ability to make glucose from substrates such as glycerol, pyruvate, and the amino acids glutamine and alanine. This provides an alternate energy

source and occurs primarily during fasting and starvation.

- **Formation of chemical compounds:** Excess glucose can also be converted into other chemical compounds (pyruvic acid, lactic acid, acetyl CoA) that can be used in metabolic pathways such as the citric acid cycle.

Lipids

Most lipids are packaged into **chylomicrons** in the enterocytes (see Chapter 25). The chylomicrons enter the lymph lacteals in the small intestine, entering the systemic blood in the large vessels in the thoracic cavity. Thus the first entry of absorbed lipid into the liver is from the systemic, not portal, circulation. Liver lipid metabolism includes the following:

- **β -Oxidation of fatty acids:** Although many tissues use β -oxidation as a source of energy when needed, the rate is very high in the liver.
- **Formation of most lipoproteins:** Very low density lipoprotein (VLDL), low density lipoprotein (LDL), and high density lipoprotein (HDL) are formed in the liver. VLDL and LDL transport TG and cholesterol to tissues. LDL is implicated in development of cardiovascular disease because it is incorporated into atherosclerotic plaques. HDL transports lipids from tissues to the liver and is considered beneficial.
- **Synthesis of cholesterol and phospholipids:** These substrates are necessary for making membranes, and cholesterol is also the precursor for steroid hormones and bile. Because of these important functions, the liver ensures a supply of these substrates by forming them from other lipids. A liver enzyme, hydroxymethylglutaryl-coenzyme A (HMG-CoA) reductase, is the rate-limiting step in cholesterol synthesis, and pharmacologic intervention by **statins** (cholesterol-lowering drugs) inhibits this enzyme.
- **Conversion of unused glycogen to TGs:** If liver glycogen is not used, it is converted to TGs, which are transported to adipose tissues in VLDLs.

Proteins

Protein metabolism in the liver is essential for survival; the liver processes dietary amino acids and participates in the processing of nitrogen wastes for excretion. The major functions of the liver in this regard are:

- **Deamination of amino acids:** This prevents toxin buildup, because aminotransferases remove the amino group from the amino acids, creating ammonia (NH_3).
- **Production of urea:** The NH_3 combines with CO_2 to form urea, thereby buffering NH_3 and allowing urinary excretion.
- **Synthesis of plasma proteins:** About 90% of plasma proteins are made in the liver. This includes:
 - albumin, which contributes to oncotic pressure

- immunoglobulins, which contribute to immune functions
- fibrinogen, which is necessary for blood clotting
- **Interconversion of amino acids:** Needed amino acids are synthesized from other available amino acids.

PRODUCTION AND SECRETION OF BILE

Bile is critical for transport of lipids through the **unstirred water layer** to the enterocytes. The unstirred water layer is the area in the gut lumen that is closest to the villous lining. There is little flow near the cells, and the water and mucus create a barrier that impedes access of the hydrophobic lipids to the enterocytes (carbohydrates and proteins have no trouble passing through this area). This challenge is met by bile, which is amphipathic (has both hydrophilic and hydrophobic regions) and thus is able to move lipids across the unstirred water. Bile and lipids form **micelles**, which act like taxis to shuttle the lipids through the unstirred water layer to the enterocytes (discussed further in Chapter 25). **Bile solids** are composed of bile salts (50%), phospholipids (40%), and smaller amounts of cholesterol (~4%), bilirubin (~2%), and water and electrolytes. Secreted bile is composed of the bile solids, water, and electrolytes.

In the liver, primary **bile acids** (cholic, chenodeoxycholic acids) are synthesized in the hepatocytes from sterol rings (from cholesterol). There are several important aspects of bile synthesis and secretion:

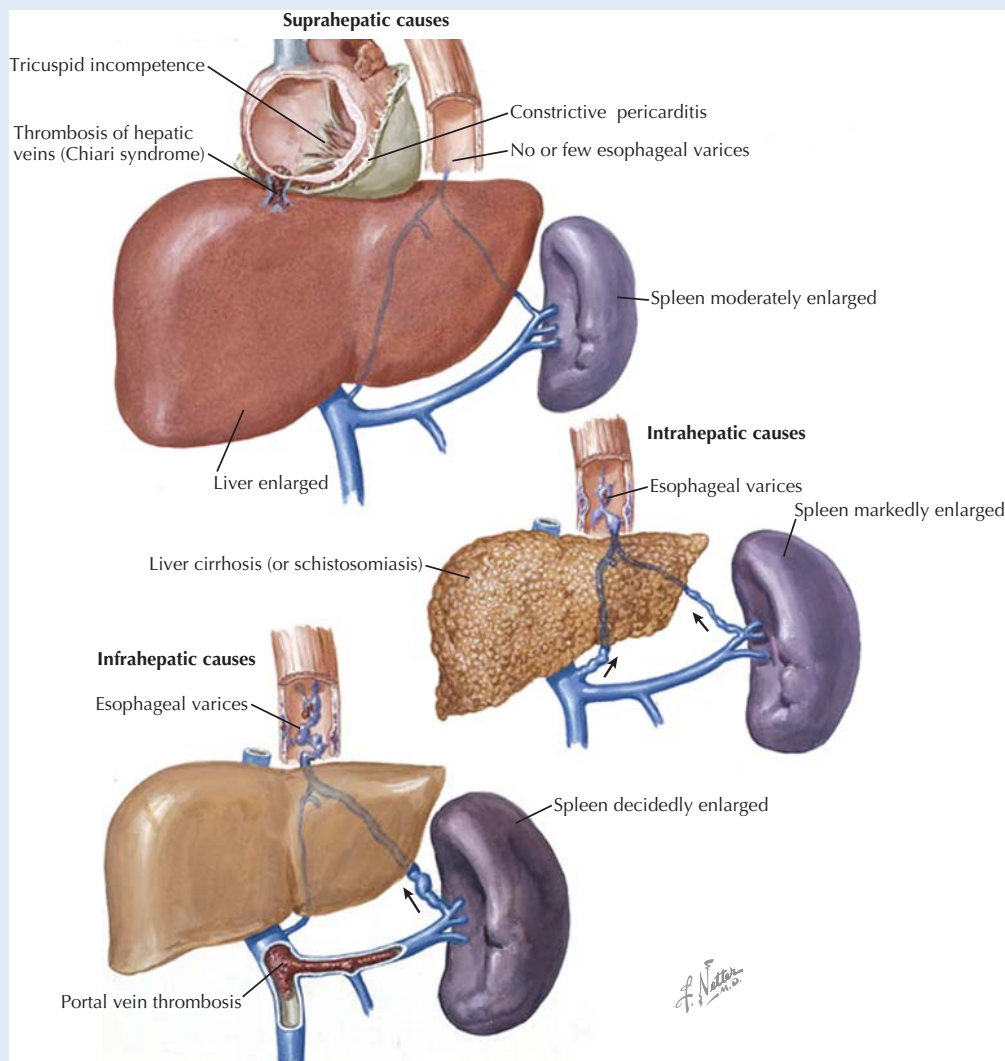
- In the liver, one side of the primary bile acid is conjugated with an amino acid (either taurine or glycine), forming a **bile salt**. This conjugation increases the water solubility of bile in the lower pH found in the duodenum. The primary bile salts are secreted into the bile canaliculi, or ductules, and into the common bile duct (see Fig. 24.2).
- Bile salts are osmotic, and their secretion will draw water, and then solvents (sodium chloride, HCO_3^-) from the cells—this is called **solvent drag** and contributes to the buffering capacity of the bile when it enters the duodenum.
- After micelles are formed and the lipids are dropped off at the enterocytes, the majority of the bile remains in the lumen of the small intestine until the terminal ileum, where Na^+ -dependent transporters recycle the primary bile into the portal vein back to the liver. This bile recycling will occur three to five times for each meal and allows for efficient absorption of the lipids without synthesis of large amounts of bile, because it is reused.
- However, with each cycle, about 10% of the bile is not absorbed but is lost in the feces. This is the major pathway by which cholesterol is removed from the system; other substances including bilirubin are also excreted. The synthesis of bile acids is under feedback control by bile salts entering the liver from the portal

CLINICAL CORRELATE**Portal Hypertension and Esophageal Varices in Obstructive Liver Disease**

Increased blood pressure in the portal vein can result from suprahepatic pathology (e.g., increase in systemic venous pressures, as with congestive heart failure) or hepatic pathology (e.g., obstructive liver disease). Obstructive liver disease is most commonly a result of cirrhosis or fibrous scarring of the liver, which severely decreases the flow of blood through the organ. In the United States, alcoholism and hepatitis C infection are the most common causes of cirrhosis. Interestingly, alcoholism causes cirrhosis of the liver or pancreatitis in individual patients, but not both diseases.

In the cirrhotic liver, the obstruction of blood flow through the liver increases pressure in the portal system. This **portal hyperten-**

sion results in increased backpressure in vessels coming from the stomach and esophagus, causing enlargement and thinning of these vessel walls, forming **varices**. The thin walls, high pressure, and increased radius in the varices make them susceptible to rupture, and because of the superficial nature of the vessels serving the esophagus, rupture may cause severe bleeding into the esophageal lumen, requiring immediate medical attention. Treatment of varices is by either sclerotherapy (injecting a solution to block the vessels), or rubber-band ligation of the varices, where a band is wound around the varices, cutting off blood flow. The atrophied area sloughs off, leaving a healed scar. Portal hypertension can also result in **ascites** and **hemorrhoids**, again caused by the backup in pressure in the portal venous system.

**Esophageal Varices**

circulation. Reduced bile salts will increase **cholesterol 7 α -hydroxylase**, which controls bile synthesis.

- In the intestines, some of the primary bile acids undergo dehydroxylation by bacteria, forming **secondary bile acids** (deoxycholic, lithocholic acids), which are less efficient at crossing the unstirred water layer, and hence are less readily absorbed.
- When the stomach is emptied and chyme is no longer present in the duodenum, the sphincter of Oddi will close. As the bile is recycled through the portal system back through the liver, it will stop at the sphincter of Oddi, and back up into the relaxed gallbladder, where it will be stored until the next meal.

BASIC ENDOCRINE FUNCTIONS

The liver produces or modifies several endocrine and paracrine substances, including the following:

- **Insulin-like growth factor-I (IGF-I)**, which is released by the liver into the circulation in response to growth hormone. IGF-I mediates many of the somatic effects of growth hormone.
- **Angiotensinogen**, which is the precursor to angiotensin I and II (angiotensin II plays an important role in blood pressure regulation).
- **Thrombopoietin (TPO)**, which stimulates stem cells in bone marrow to differentiate to megakaryocytes, which give rise to platelets. Platelets participate in blood clotting.
- **Hepatocyte growth factor (HGF)**, which acts locally to stimulate regeneration of liver cells and is especially important when the organ is damaged.
- **Vitamin D metabolism:** The liver hydroxylates cholecalciferol (from diet or synthesized in skin) to form 25-hydroxycholecalciferol. This is still inactive and must be further hydroxylated in the kidney to become the active form of vitamin D, 1,25-dihydroxycholecalciferol. Vitamin D is a key regulator of intestinal calcium absorption.

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CHAPTER 25

Digestion and Absorption

ANATOMY AND NUTRIENT ABSORPTION

Absorption of nutrients in the small intestine depends on an intact villous lining, which is designed to facilitate efficient nutrient uptake into the body. The lumen of the small intestine is composed of circular folds, villi, and microvilli (Fig. 25.1). This increases the surface area of the small intestine to approximately 250 square meters and dramatically increases the efficiency of the system so that most of the nutrient absorption will occur by mid- to late-jejunum. As a rule of thumb, ~25% to 30% of digestion occurs preduodenally, and ~70% to 75% occurs in the duodenum and jejunum.

The upper half of the villi is the site of membrane-bound enzymes (brush border enzymes) for final digestion of carbohydrates and proteins. This is also the site of absorption. The bottom part of the villi form the crypts of Lieberkühn, from which buffers and mucus are secreted, as previously discussed in Chapter 23. Finally, because there is laminar flow through the lumen, the slowest movement is near the enterocytes, and this is slowed even further because of the mucus secreted to protect the cells. This creates an **unstirred water layer** that molecules must pass through to access the enterocytes. This is no problem for movement of most nutrients but creates a problem for hydrophobic, lipid-based molecules.

CARBOHYDRATE DIGESTION AND ABSORPTION

Much of the carbohydrate in our diet is in the form of starch, sucrose (table sugar), lactose (milk sugar), and fructose. **Starch** is a large, branched, long-chain polysaccharide synthesized by plants; it represents a large portion of the carbohydrates consumed in most diets. The glucose moieties within the starch molecule are bound by α -1,4 glycosidic linkages, with α -1,6 linkages at branch points. **Sucrose** and **lactose** are **disaccharides**. We also ingest large quantities of the monosaccharide **fructose**, which is present in fruits and many foods using “high-fructose corn syrup” as a sweetener.

Although we ingest long-chain sugars and oligosaccharides, we can only absorb monosaccharides. Carbohydrate digestion begins in the mouth and continues through to the small intestine (Fig. 25.2):



Starch, glycogen, and cellulose are all polymers of glucose. Starch is a storage form of glucose in plants and is a major source of energy in our diet. Although animals store glucose in their muscles in the form of glycogen, the meat we consume contains little or no glycogen, because it is broken down after animals are slaughtered. Cellulose is the structural component of the cell walls of green plants and has no caloric value to humans. Cellulose is plant fiber and is composed of glucose molecules bound by β -1,4 linkages in large, straight chains. Humans do not have an enzyme that can digest β -linkages, so this “fiber” remains in the gastrointestinal (GI) lumen. Fibers attract water, loosening stools and adding bulk. High-fiber diets have other health-related advantages, including a blood cholesterol-lowering effect and enhanced glucose tolerance (antidiabetogenic effect). In contrast to other mammals, ruminants such as cows are able to digest cellulose due to the presence of symbiotic prokaryotic organisms in their GI tracts.

- **Mouth:** Salivary α -amylase begins starch digestion, breaking α -1,4 linkages and creating smaller oligosaccharides (maltose and isomaltose, both disaccharides consisting of two linked glucose molecules, as well as larger oligosaccharides and polysaccharides). It is inactivated by the acid pH in the stomach.
- **Small intestine:** Pancreatic α -amylase continues the digestion, forming more maltose and isomaltose.
- **Small intestine brush border:** As the chyme contacts the villous brush border, specific **brush border saccharidases** digest the maltose, isomaltose, sucrose, and lactose to their constitutive monosaccharides (Table 25.1).

In the small intestine, glucose and galactose are transported into the enterocytes with sodium via SGLT-1 transporters (secondary active transport); fructose has its own transport protein, GLUT-5 (through facilitated transport). The monosaccharides leave the basolateral membranes by facilitated transport through GLUT-2, diffuse through the interstitial space into the capillary, and are transported through the portal vein to the liver for processing and release into the systemic circulation. Carbohydrates are efficiently digested and quickly absorbed.

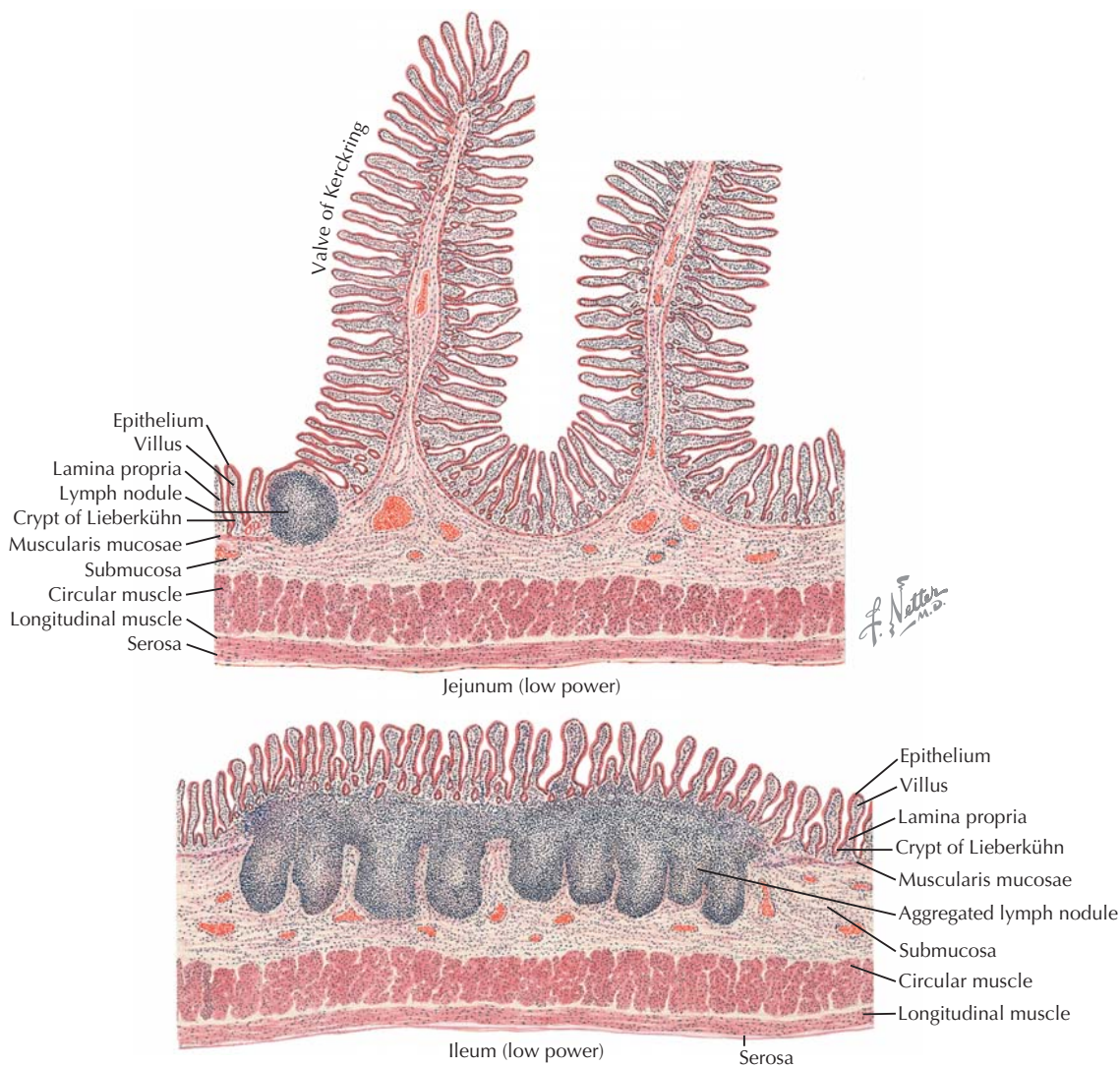


Figure 25.1 Small Intestine Surface Area The small intestine has circular folds, villi, and microvilli (hairlike projections on villi), which increase the surface area for absorption to about 600 times the area of a straight tube. This allows efficient absorption of nutrients. The upper half of the villi are the site of the brush border saccharidases and proteases (for final digestion of carbohydrates and proteins), and the upper half is also where absorption occurs.

Table 25.1 Brush Border Saccharidases and Products

Brush Border Enzyme	Substrate	Product(s)
Maltase	Maltose	Glucose
Isomaltase	Isomaltose	Glucose
Sucrase	Sucrose	Glucose + Fructose
Lactase	Lactose	Glucose + Galactose



Because glucose is absorbed into the enterocytes by SGLT-1 transporters, the rapid absorption of glucose facilitates sodium absorption, and hence chloride and water absorption. This is the principle behind **oral rehydration therapy (ORT)** for dehydration or enteric diseases such as cholera.

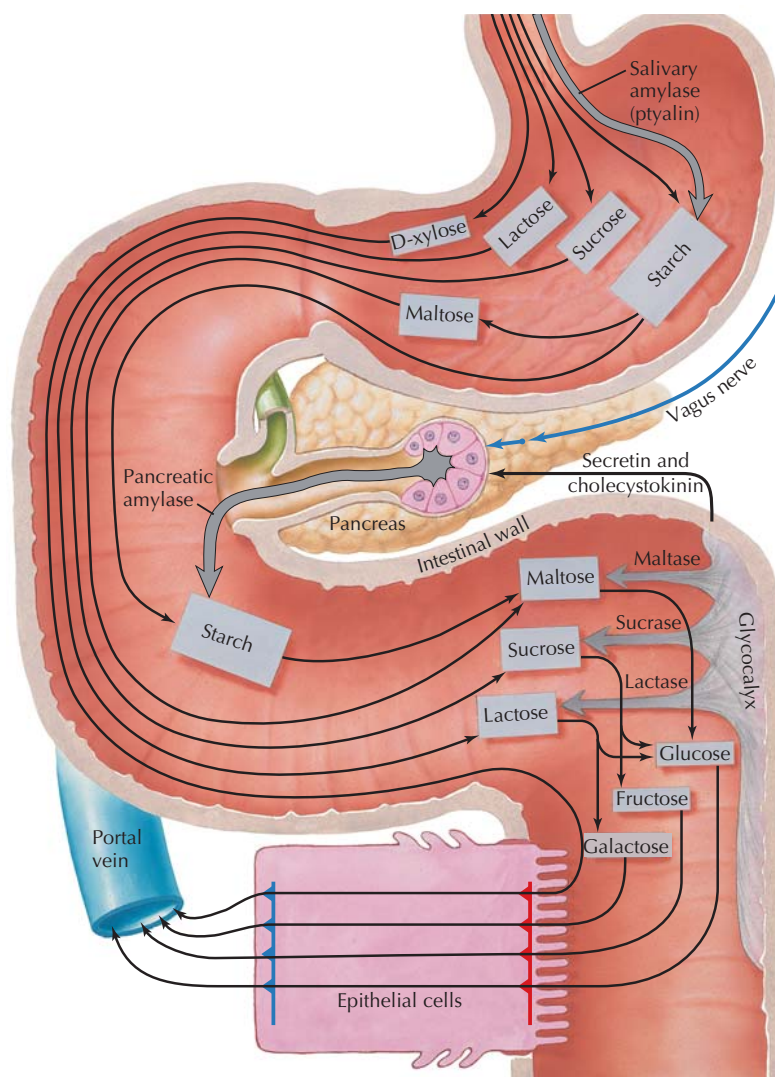


Figure 25.2 Carbohydrate Digestion and Absorption Starch digestion begins in the mouth with salivary α -amylase, which produces smaller malto-oligosaccharides. In the small intestine lumen, pancreatic α -amylase continues to digest the starches. At the same time, brush border saccharidases perform the final digestion of maltose and isomaltose (by maltase and isomaltase, producing glucose molecules), and the disaccharides sucrose (by sucrase, producing glucose and fructose) and lactose (by lactase, producing glucose and galactose). The monosaccharides then are transported into the enterocytes by secondary active transport with sodium (glucose and galactose) and facilitated transport (fructose; not shown on diagram).

PROTEIN DIGESTION AND ABSORPTION

Dietary protein digestion begins in the stomach and continues in the small intestine:

- **Stomach:** Pepsinogens are secreted from the gastric chief cells and are activated to pepsins by stomach acid. Pepsins are endopeptidases (such as trypsin and chymotrypsin) and hydrolyze inner peptide bonds, creating smaller oligopeptides. Pepsins are inactivated in the higher pH of the duodenum.

- **Small intestine:** Pancreatic proteases are responsible for digesting the oligopeptides down to smaller peptides. As stated in Chapter 23, the pancreatic proteases are released as zymogens into the duodenum primarily in response to cholecystokinin (CCK). The brush border enzyme **enterokinase** activates trypsinogen to **trypsin**, and the trypsin activates the other endopeptidases (chymotrypsin, elastase) and the exopeptidases (carboxypeptidase A and carboxypeptidase B). Once activated, trypsin can also activate trypsinogen. The pancreatic proteases

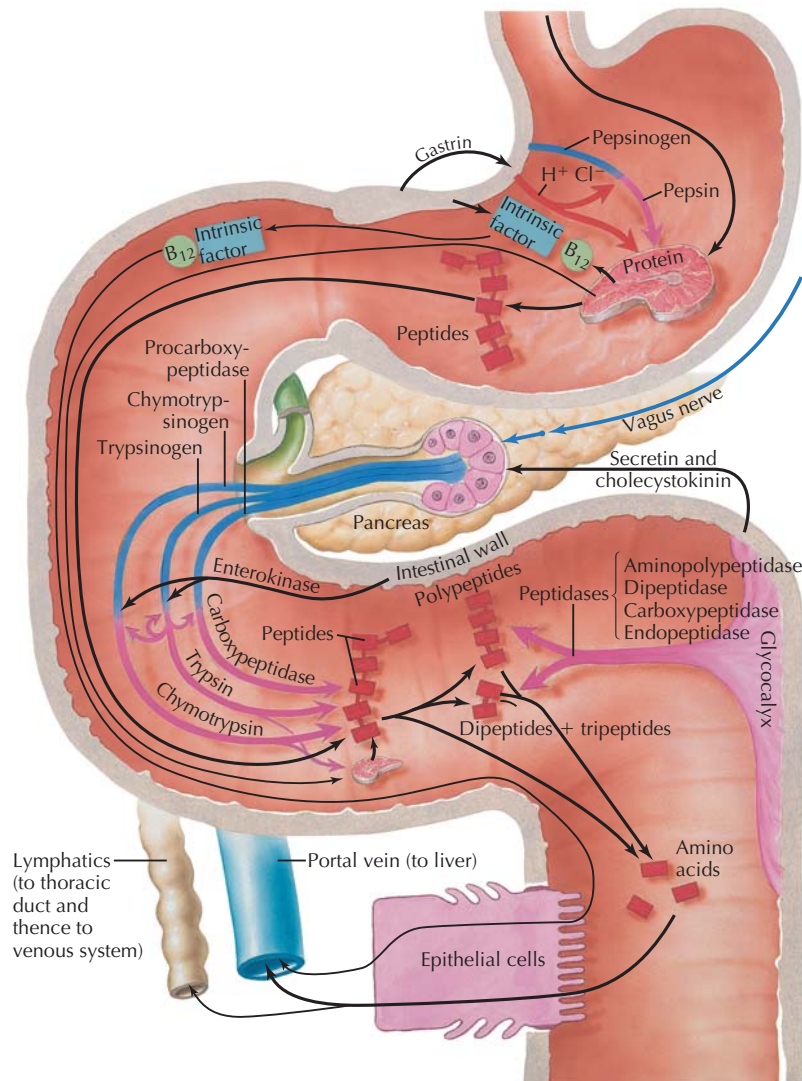


Figure 25.3 Protein Digestion and Absorption Protein digestion begins in the stomach with HCl and pepsins, which break protein down to smaller polypeptides. In the lumen of the small intestine, pancreatic proteases (trypsin, chymotrypsin, carboxypeptidase) are activated and continue the digestion to smaller peptide chains. Final digestion occurs by the brush border proteases, which produce dipeptides, tripeptides, and single amino acids. These molecules are transported into the cells via secondary active transport with sodium (amino acids), or coupled with H^+ (dipeptides and tripeptides). In the cells, the dipeptides and tripeptides are digested to amino acids by cytoplasmic peptidases.

continue hydrolysis of peptide bonds, making smaller oligopeptides (Fig. 25.3).

- **Small intestine brush border:** A variety of brush border peptidases hydrolyze the peptides to amino acids and dipeptides and tripeptides, which can then be absorbed.
- **Enterocytes:** Cytoplasmic peptidases digest the dipeptides and tripeptides to amino acids, which can exit the cells.

In the small intestine, most of the proteins are absorbed into the enterocytes in dipeptide and tripeptide form, via H^+



As previously stated, storing and secreting the pancreatic proteolytic enzymes as zymogens helps prevent digestion of the pancreatic tissue. However, because trypsin can autoactivate itself, the pancreas also produces **trypsin inhibitor**, which deactivates trypsin in the pancreas and ducts. In the small intestine, the large amount of active trypsin is also prevented from damaging the intestinal mucosa by the presence of other trypsin inhibitors (PSTI, pancreatic secretory trypsin inhibitors), which can compete with proteins for trypsin. The PSTI have also been shown to stimulate growth of intestinal epithelial cells.

symporters that are specific for the peptides. Amino acids have different Na^+ -dependent transporters for basic, acidic, neutral, and imino acids. Once inside the intestinal cells, the cytoplasmic peptidases hydrolyze the dipeptides and tripeptides to amino acids, which leave the cells by facilitated transport into the capillaries. A small amount of the dipeptides and tripeptides can be transported through the cell and into the blood, but the mechanism is unclear.

LIPID DIGESTION AND ABSORPTION

Almost all (98%) of dietary lipids are triglycerides (TG), with the remainder cholesterol esters and phospholipids (PL). Lipids are easily hydrolyzed to molecules that can be absorbed; however, their hydrophobicity does not allow easy access to the absorptive cells of the small intestine brush border. As a result, there is a complex mechanism for efficiently moving lipids through the unstirred water layer to the enterocytes. It should be noted that although there is not a lot of lipid digestion in the upper GI tract in adults, lingual and gastric lipases do have a role in hydrolysis of lipids in the neonate:

- **Mouth:** Lingual lipase is secreted from Von Ebner's glands of the tongue into the saliva and begins hydrolysis of TGs to diglycerides and free fatty acids (FFA). The enzyme remains active in the stomach.
- **Stomach:** Gastric lipase is secreted from the chief cells of the gastric pits and also hydrolyzes TGs to diglycerides and FFA. Again, in the adult this appears to be a minor factor in lipid digestion.
- **Small intestine:** Various lipases are secreted in active form from the pancreas in response to the action of CCK on the pancreatic acinar cells. **Pancreatic lipase** hydrolyzes TGs to monoglycerides (MG) and FFA; **procolipase** is also secreted and is activated in the lumen of the duodenum by trypsin, forming colipase, which facilitates the enzymatic action of pancreatic lipase; **cholesterol ester hydrolase** hydrolyzes cholesterol esters to cholesterol and FFA; **phospholipase A₂** hydrolyzes PL to lysophospholipids and FFA. These cleavage products are all able to diffuse into the enterocytes; however, they must first be incorporated into micelles to traverse the unstirred water layer.

Micelle Formation

Micelles are aggregates of bile salts and small lipids. The polar (hydrophilic) ends of the bile are oriented outward, and the hydrophobic core allows incorporation of small lipids. The micelles function to move lipids through the unstirred water layer, allowing lipids access to the enterocytes.

Most lipids are in the last portion of chyme entering the small intestine. In the duodenum, the lipids are surrounded by bile

and exposed to pancreatic lipases. However, although the pancreatic lipase can readily hydrolyze triglycerides, lipase access to lipids is blocked by bile. The colipase is critical for lipase action. It displaces the bile from the lipids, allowing the lipase to hydrolyze the lipids.

These are the steps for micelle formation (Fig. 25.4):

1. The detergent action of bile emulsifies the fat globules, forming smaller fat droplets, which increases the surface area for digestion.
2. Colipase displaces the bile from the lipids.
3. Pancreatic lipase binds to colipase and hydrolyzes the TGs to monoglycerides and FFA. Cholesterol ester hydrolase and phospholipase A₂ hydrolyze cholesterol esters and phospholipids (these enzymes are not affected by bile).
4. When there is a critical amount of bile and digested lipids, they form a **micelle**, with the hydrophilic ends of bile salts on the outside, and lipophilic ends inside surrounding the lipid products.
5. The micelles diffuse through the unstirred water layer to the enterocytes, and the lipids are released from the micelle and diffuse through the enterocyte membrane, leaving the bile salts in the lumen.
6. The bile salts continue down the small intestine to the terminal ileum, where most primary bile salts enter the intestinal cells through Na^+ -dependent transporters. The bile diffuses out of the cell into the portal circulation back to the liver.

Intracellular Lipid Processing

Once the lipids diffuse into the enterocytes, they are re-esterified with FFA in the smooth endoplasmic reticulum to reform triglycerides, cholesterol esters, and phospholipids (see Fig. 25.4). The lipids are surrounded by a β -lipoprotein (ApoB) coat forming **chylomicrons**. The chylomicrons are packaged into secretory vesicles and leave the cell by exocytosis. The chylomicrons enter the lymph lacteals; they are too large to enter capillaries. They enter the systemic blood in the large vessels in the thorax, and return to the liver for processing through the systemic circulation. A small amount of short-chain fatty acids are not incorporated into chylomicrons and are able to diffuse into the portal blood to the liver. This is a minor pathway.



β -Lipoprotein is critical for chylomicron formation. When it is absent, lipids will not be able to exit the intestinal cells and enter the lymph. Fat builds up in the enterocytes. As the villous cells are replaced and sloughed off into the lumen, the lipid is excreted. This condition of **abetalipoproteinemia** results in the inability to absorb lipids.

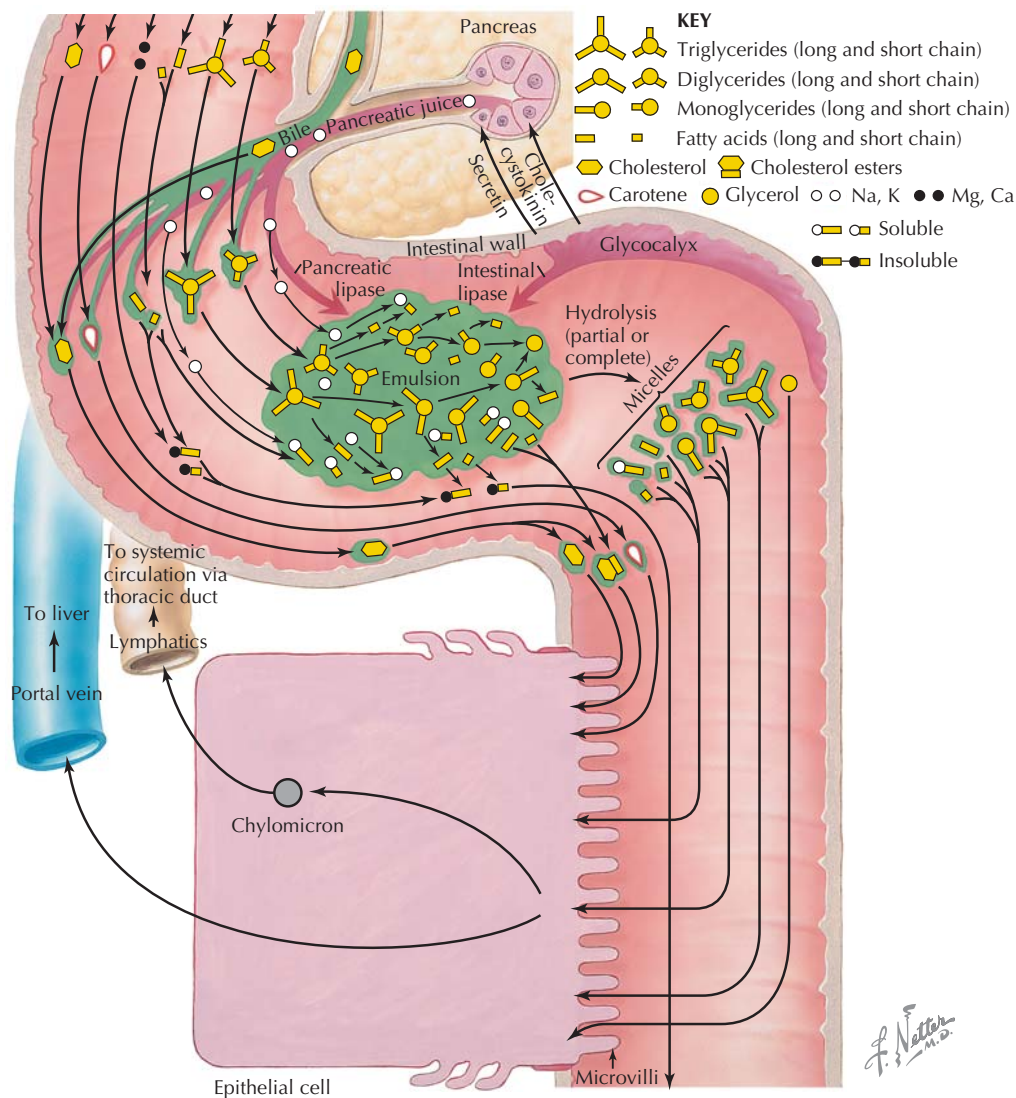


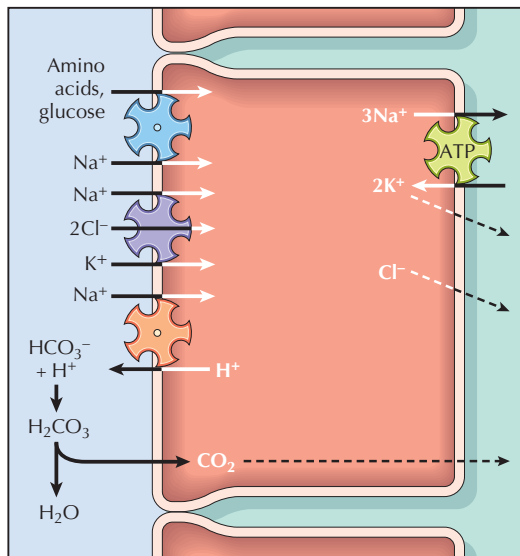
Figure 25.4 Lipid Digestion and Absorption Although some lipid digestion occurs preduodenally by lingual and gastric lipases, most digestion occurs in the lumen of the small intestine by pancreatic lipase. The overall process of digestion and absorption is complex, because the lipids must first be incorporated with bile into micelles and shuttled through the unstirred water layer to the enterocytes. The lipids diffuse into the enterocytes and are re-esterified with free fatty acids in the smooth endoplasmic reticulum and packaged into chylomicrons for transport into the lymph, eventually passing through the thoracic duct into venous circulation. Some soluble fats (glycerol and short-chain and medium-chain fatty acids) can access the enterocytes without micelles and are small enough to be taken up into the portal circulation.

ELECTROLYTE AND WATER ABSORPTION

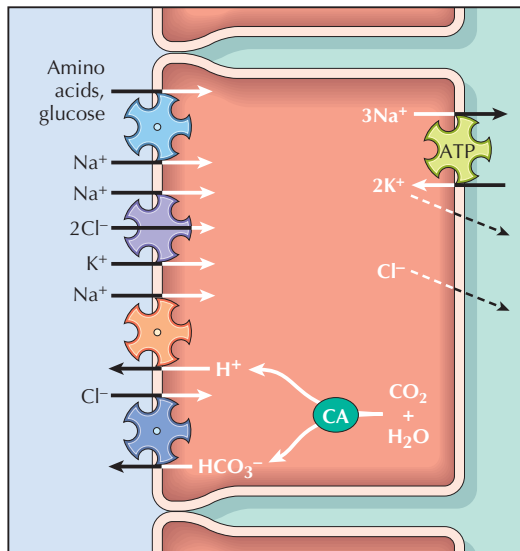
In addition to the ~2 liters (L) of fluid ingested each day, approximately 7 or more additional liters are added to the GI tract in various segments to facilitate digestion and absorption of the nutrients. Absorption of the 9 L of fluid occurs as nutrients and electrolytes are absorbed. Within the small intestine, several mechanisms of absorption occur in various segments (Fig. 25.5):

- **Jejunum:** The large surface area in the small intestine makes digestion and absorption extremely efficient. The bulk of the nutrients, fluid, and electrolytes is absorbed by mid- to late-jejunum, although there is significant electrolyte and fluid absorption in the ileum.
- **Sodium** absorption is a driving force for nutrient and water absorption, and the basolateral Na^+/K^+ ATPase

A. Jejunum



B. Ileum



C. Colon

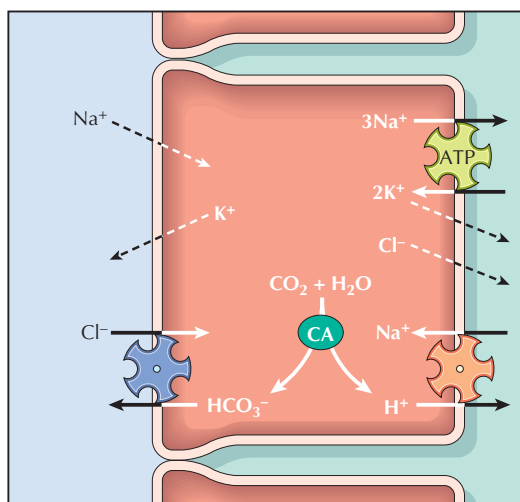

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Figure 25.5 Electrolyte and Fluid Transport A, Sodium absorption in the jejunum and ileum occurs through a variety of mechanisms, with sodium movement facilitating entry of other molecules such as glucose, galactose, amino acids, and chloride. This also creates an osmotic gradient for water absorption. B, The ileum secretes HCO_3^- into the gut lumen in exchange for Cl^- , and the Cl^- exits through the basolateral membrane by facilitated transport. C, The colon is sensitive to the effects of aldosterone, which will enhance sodium absorption and potassium secretion. The water follows the osmotic gradient, dehydrating the chyme and producing feces.

maintains the low intracellular concentration necessary for the luminal Na^+ to enter the cells.

- **Bicarbonate** is formed by the intracellular CO_2 metabolism and exits the cell through the basolateral membrane by facilitated transport. The **water** follows the osmotic gradient created by the electrolyte movement (see Fig. 25.5A).
- **Ileum**: As noted, the Na^+ -dependent transporters for sugars and amino acids are also present in the ileum, and this redundancy helps ensure nutrient absorption if absorption is reduced in the jejunum. **Sodium** is absorbed in the same manner as in the jejunum; however, in the ileum and colon, the bicarbonate formed in the cells is *secreted* into the lumen via $\text{HCO}_3^-/\text{Cl}^-$ antiporters, and the **chloride** leaves the basolateral side via facilitated transport (see Fig. 25.5B).
- **Colon**: Aldosterone stimulates luminal sodium absorption by Na^+/K^+ and Na^+/H^+ exchangers. As water follows the sodium, the chyme is dehydrated, producing feces. The colon usually absorbs ~400 to 500 milliliters of water per day, which follows sodium absorption, and water absorption in the colon can increase to ~1 L per day, when aldosterone is elevated (see Fig. 25.5C).

DIVALENT CATION ABSORPTION

Calcium and iron are absorbed early in the small intestine, from the duodenum through early- to mid-jejunum (first half of the jejunum). Because the early part of the small intestine has a lower pH, it favors the reduced (Fe^{2+}) form of iron and keeps the cations from forming insoluble salts.

Calcium

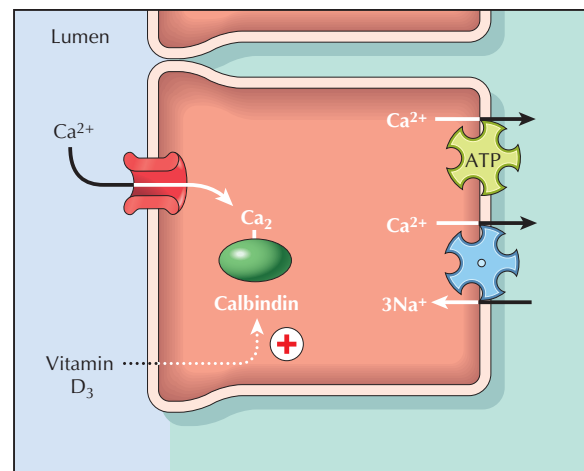
Calcium absorption is regulated by active **vitamin D** (1,25-dihydroxycholecalciferol), which increases the calcium channels (TRPV-6) at the luminal membrane, as well as the cytosolic binding protein **calbindin**. Calbindin binds Ca^{2+} , keeping the intracellular free Ca^{2+} levels very low (10^{-6} M), maintaining a gradient for Ca^{2+} entry into the cell, and allowing Ca^{2+} -dependent messenger systems to function properly. The Ca^{2+} exits the basolateral side through active Ca^{2+} ATPase pumps and $\text{Na}^+/\text{Ca}^{2+}$ antiporters (Fig. 25.6A). The activity of the Ca^{2+} ATPase is also increased by vitamin D.

Iron

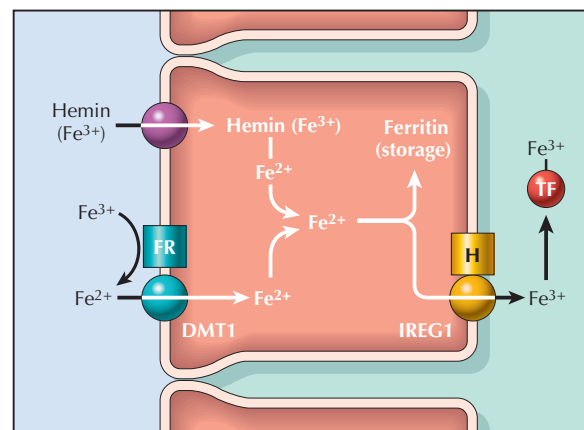
Free iron is toxic to cells, and in the body iron is always bound to proteins. Ingested iron is either in organic (hemoglobin or myoglobin from meats) or free (from vegetables) forms.

Organic forms can diffuse into the enterocytes, and intracellular xanthine oxidase frees the iron, which is then bound in ferritin in the cell or is transported out of the cell into the blood, where it is bound to transferrin (TF).

A. Ca^{2+}



B. Iron



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Figure 25.6 Calcium and Iron Absorption A, Calcium entering the enterocytes through the TRPV-6 Ca^{2+} channel is bound by cytosolic calbindin. Transport out of the cell is via Ca^{2+} ATPase pumps and $\text{Na}^+/\text{Ca}^{2+}$ exchangers. Active vitamin D enhances Ca^{2+} absorption by increasing calbindin production and Ca^{2+} pump activity. B, Ingested iron is in both organic (heme) and inorganic forms. Heme can enter enterocytes, and intracellular heme oxygenase releases the iron (Fe^{2+}), which is then bound in ferritin stores or shuttled out of the cell through the transmembrane protein hephaestin (H) and bound to transferrin (TF) in the blood. It is transported to the liver for storage, or the bone marrow for hemoglobin and red blood cell production. Inorganic iron in the ferric form (Fe^{3+}) can be reduced to Fe^{2+} by ferrireductase (FR) in the luminal membrane, and then the Fe^{2+} can enter the enterocytes through the divalent metal transporter 1 (DMT-1), and be handled as described above.

The luminal membranes in the jejunum have divalent metal transport proteins (DMT-1), which transport free Fe^{2+} into the cell, where it is bound in ferritin. The iron can also be transported out of the cell into the blood, where it is bound to transferrin (TF) (see Fig. 25.6B).

In the blood, the transferrin-bound iron is transported to the bone marrow for hemoglobin and red blood cell synthesis, or

Table 25.2 Vitamin Absorption

Vitamin	Site of Absorption	Mechanism
WATER SOLUBLE VITAMINS		
Vitamin C	Ileum	Na ⁺ -coupled/2° active
Thiamin (B1)	Jejunum	Na ⁺ -coupled/2° active
Riboflavin (B2)	Jejunum	Na ⁺ -coupled/2° active
Biotin	Jejunum	Na ⁺ -coupled/2° active
Vitamin B12	Ileum	Facilitated diffusion
Pyridoxine (B6)	Jejunum and ileum	Passive diffusion
FAT-SOLUBLE VITAMINS		
Vitamin A	Jejunum and ileum	Passive diffusion
Vitamin D	Jejunum and ileum	Passive diffusion
Vitamin E	Jejunum and ileum	Passive diffusion
Vitamin K	Jejunum and ileum	Passive diffusion

(Reprinted with permission from Hansen J: Netter's Atlas of Human Physiology, Philadelphia, Elsevier, 2002.)

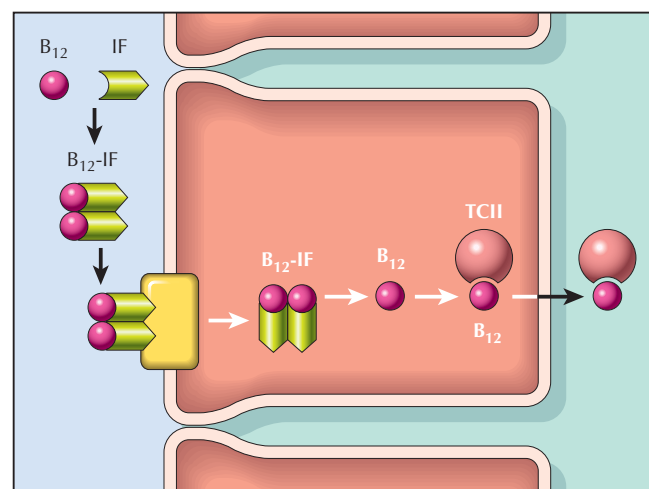
to the liver and spleen for storage. Typically, ~1 milligram (mg) of iron is absorbed daily, which balances the ~1 mg loss of iron (from intracellular ferritin stores) that occurs when enterocytes slough off each day. If iron utilization increases (such as in growing children and menstruating women), iron intake must increase, or anemia will result.

VITAMIN ABSORPTION

Fat soluble vitamins A, D, E, and K are transported in micelles to the enterocytes, although vitamin A and D are less hydrophobic, and are also able to access the enterocytes independently. Table 25.2 details the sites and mechanism of vitamin absorption in the small intestine.

Vitamin B12 Absorption

Vitamin B12 (cobalamin) is an essential vitamin and is protected from digestion until it is absorbed in the terminal ileum. The process of protecting B12 begins in the mouth:


Vitamin B₁₂

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Figure 25.7 Vitamin B12 Absorption Vitamin B12 (cobalamin) is protected from digestion in the small intestine by binding to intrinsic factor (IF) released from the gastric parietal cells. B12/IF dimers are recognized by receptors on the enterocytes in the terminal ileum, and the complex is internalized. The B12 is bound to transcobalamin II (TCII) for transport through the blood to the liver for storage, or bone marrow for red blood cell maturation.

- In the saliva, vitamin B12 is bound by **R-proteins**.
- **Intrinsic factor** is secreted by gastric parietal cells. It does not bind B12 until it reaches the duodenum, where the R-proteins are cleaved by trypsin.
- The **B12-intrinsic factor complex dimerizes** and continues down the small intestine to the terminal ileum, where the complex binds to a transport protein and enters the cell. The B12 binds to **transcobalamin II** in the cytosol, exits the cell, and is transported to the bone marrow (where it promotes red blood cell maturation) or to the liver for storage (Fig. 25.7).



Surgical removal of the **terminal ileum** causes multiple problems, including loss of both bile recycling and vitamin B12 absorption. In this circumstance, bile production by the liver is upregulated, but production is not sufficient for absorption of fat contained in a normal meal, and some steatorrhea may result. To treat vitamin B12 deficiency, injections are administered every few months (infrequent injections are sufficient because excess B12 will be stored in the liver).

CLINICAL CORRELATE**Gluten Enteropathy and Celiac Disease**

Celiac sprue is an autoimmune “allergic” reaction to **gluten proteins** in wheat and results from the plasma cell production of antibodies (IgA and IgG) against the intestinal epithelium. The α -gliadin (part of gluten proteins) is deaminated in the enterocytes, increasing the immunogenicity of gliadin. The antibodies cross-react with the intestinal tissue, producing inflammation; this results in flattening of the villous lining and crypt hyperplasia. The surface area for absorption is diminished, and brush border enzyme activity is impaired. The loss of brush border enzymes with villous atrophy will reduce final digestion of carbohydrates

and protein and cause general malabsorption of micronutrients and macronutrients. The presence of undigested nutrients lower in the GI tract facilitates bacterial production of gasses, causing cramps, odiferous gas, and feces. Diarrhea, weight loss, anemia, and vitamin deficiency can occur.

Although it is suggested that 1 in 250 persons have some degree of gluten enteropathy, most cases are considered “silent” sprue, with little or no GI symptoms. In addition to GI complications, the immune reaction to wheat can also cause respiratory (rhinitis, asthma) and skin (eczema, dermatitis herpetiformis, and/or hives) problems. Primary treatment involves complete avoidance of gluten in the diet.



Gluten Enteropathy (Celiac Disease)

Review Questions

CHAPTER 21: OVERVIEW OF THE GASTROINTESTINAL TRACT

1. All of the following are functions of the gastrointestinal tract EXCEPT:

- A. secretion of endocrine hormones.
- B. secretion of digestive enzymes.
- C. absorption of nutrients from chyme.
- D. regulation of systemic blood flow.
- E. storage of chyme.

2. Motility and secretion through the gastrointestinal tract is regulated by all of the following factors EXCEPT:

- A. duodenal hormones.
- B. chemoreceptors.
- C. autonomic nerves.
- D. enteric nerves.
- E. growth hormone.

3. In the absence of *extrinsic* innervation to the gastrointestinal tract,

- A. propulsive movements would not occur in the small intestine.
- B. propulsive movements would occur in the small intestine, but secretion into the small intestine would be abolished.
- C. secretion would be abolished in the entire GI tract.
- D. motility and secretion throughout the tract would be stimulated by local mechanoreceptors, chemoreceptors, and osmoreceptors, as well as hormones.
- E. motility and secretion would be affected only in the large intestine.

4. Input from vagal efferent nerves to the gastrointestinal tract regulates the following actions EXCEPT:

- A. lower intestinal motility.
- B. primary esophageal peristalsis.
- C. the cephalic phase of gastric acid secretion.
- D. receptive relaxation in the stomach.
- E. stimulation of hepatic bile production.

CHAPTER 22: MOTILITY THROUGH THE GASTROINTESTINAL TRACT

5. Action potentials in the GI smooth muscle:

- A. occur only in the upper GI tract.
- B. are stimulated when the slow waves are depolarized above -40 mV.

- C. occur only in the lower GI tract.
- D. result from influx of sodium ions into the smooth muscle cells.
- E. are only stimulated by extrinsic nerves.

6. Slow waves constituting the basal electrical rhythm:

- A. are hyperpolarized by stretch, ACh, and gastrin.
- B. occur at a consistent rate throughout the GI tract.
- C. are undulations in the resting membrane potential resulting from Na^+/K^+ ATPase activity.
- D. are depolarized by sympathetic nerve stimulation.
- E. are absent in the colon.

7. In the small intestine, peristalsis:

- A. occurs primarily in response to the chemical composition in the chyme.
- B. proceeds in both forward and backward directions from the site of contraction.
- C. involves receptive relaxation of the muscle proximal (toward the mouth) to the bolus of chyme.
- D. involves contraction of the muscle proximal to the bolus of chyme.
- E. is dependent on parasympathetic innervation.

8. Select the FALSE answer about the migrating myoelectric complex (MMC).

- A. The MMC is a series of contractions that “sweep” undigested fibers and bacteria into the colon.
- B. The parasympathetic nervous system controls the MMC contractions.
- C. In each cycle, the main contractions of the MMC occur during phase III.
- D. The hormone motilin stimulates phase III contractions.
- E. The MMC occurs during fasting, and stops upon feeding.

CHAPTER 23: GASTROINTESTINAL SECRETIONS

9. Secretions entering the lumen of the stomach include all of the following substances EXCEPT:

- A. HCl.
- B. lipase.
- C. mucus.
- D. gastric inhibitory peptide.
- E. intrinsic factor.

10. Parietal cell gastric acid secretion is regulated:

- A. directly by secretin.
- B. directly by peptide YY.
- C. directly and indirectly by the vagus nerve.
- D. indirectly by somatostatin.
- E. indirectly by gastrin.

11. Select the TRUE statement about gastric acid secretion.

- A. Histamine suppresses acid secretion.
- B. Blocking the H^+/K^+ ATPase pump increases alkaline tide.
- C. Basolateral HCO_3^-/Cl^- exchangers decrease intracellular Cl^- in parietal cells.
- D. Parasympathetic stimulation of acid secretion is limited to the cephalic phase of acid secretion.
- E. Proton pump (H^+/K^+ ATPase) activity is the rate-limiting step in acid secretion.

12. Which of the following is NOT a mechanism by which sodium ions enter the intestinal epithelial cells from the luminal membrane?

- A. Diffusion down the sodium concentration gradient
- B. Active transport via Na^+/K^+ ATPase
- C. Co-transport with certain amino acids
- D. Co-transport with monosaccharides
- E. In exchange for H^+

13. Select the TRUE statement about pancreatic secretions.

- A. The release of pancreatic enzymes into the duodenum is primarily stimulated by gastrin.
- B. The release of pancreatic enzymes into the duodenum is primarily stimulated by the sympathetic nerves.
- C. Secretin is released in response to low duodenal pH and acts at the pancreas to stimulate the release of electrolyte buffer solution.
- D. Pancreatic proteases are released in active form.
- E. Pancreatic lipase digests all lipids.

CHAPTER 24: HEPATOBILIARY FUNCTION

14. The liver performs all of the following functions EXCEPT:

- A. synthesis of cholesterol.
- B. vitamin production.
- C. β -oxidation of fatty acids.
- D. bile acid production.
- E. metabolism of proteins.

15. Obstruction of blood flow through the liver (from cirrhosis or hepatitis) will

- A. have no effect on gastrointestinal digestion or absorption.

- B. have no effect on bile secretion.
- C. increase bile secretion.
- D. increase portal vein pressure.
- E. reduce hepatic blood content.

16. Metabolism of lipids within the hepatocytes includes:

- A. formation of most lipoproteins, including very low and low density lipoproteins.
- B. net breakdown of phospholipids.
- C. net breakdown of cholesterol.
- D. formation of steroid hormones.
- E. None of the above.

17. Bile salts:

- A. hydrolyze dietary fats.
- B. are essential for creating micelles.
- C. are efficiently absorbed into the portal blood all along the small intestine.
- D. do not contribute to intestinal buffering of acidic chyme.
- E. production is not affected by loss of the terminal ileum.

CHAPTER 25: DIGESTION AND ABSORPTION

18. Which of the following would delay or diminish the absorption of proteins?

- A. Inhibition of gastric somatostatin release
- B. An increase in Na^+/K^+ ATPase activity in the basolateral membranes of the enterocytes
- C. Stimulation of gastric pepsinogen release
- D. pH of 3 in the duodenum and jejunum
- E. Secretion of cholecystokinin

19. Select the TRUE statement about carbohydrate digestion.

- A. Digestion of carbohydrates begins in the intestines.
- B. Loss of the intestinal brush border has no effect on carbohydrate digestion.
- C. Disaccharidases are secreted into the duodenum from the crypts of Lieberkühn.
- D. Starch digestion is initiated by α -amylase.
- E. The majority of carbohydrate digestion occurs before the duodenum.

20. When chyme is present in the stomach and duodenum, all of the following actions occur EXCEPT:

- A. stimulation of intestinal buffers by gastrin.
- B. stimulation of hepatic bile production by vagal afferents.
- C. stimulation of pancreatic enzyme secretion by cholecystokinin.
- D. stimulation of intestinal buffer secretion by secretin.
- E. conversion of trypsinogen to trypsin by enterokinase.

21. In a patient with celiac sprue,
- A. gastric ulcerations are common.
 - B. fibrosis decreases intestinal absorption and reduces portal blood flow.
 - C. flattening of intestinal villi decreases enzymatic digestion and reduces the area for absorption.
 - D. hypersecretion of mucus obstructs absorption.
 - E. hyposecretion of mucus allows acidification of the intestinal lumen, decreasing enzymatic digestion.
22. The absorption of vitamin B12 (cobalamin) is dependent on all of the following EXCEPT:
- A. the integrity of the terminal ileum.
 - B. the concentration of B12 in the enterocytes.
 - C. the presence of R-protein in saliva.
 - D. the presence of intrinsic factor in gastric juice.
 - E. the presence of vitamin B12/intrinsic factor binding sites in the ileum.